

BUILDING  
INFORMATION  
MANAGEMENT  
CAPSTONE PROJECT  
2024

# BIM PORTFOLIO

Presented by

Vivek Khanna

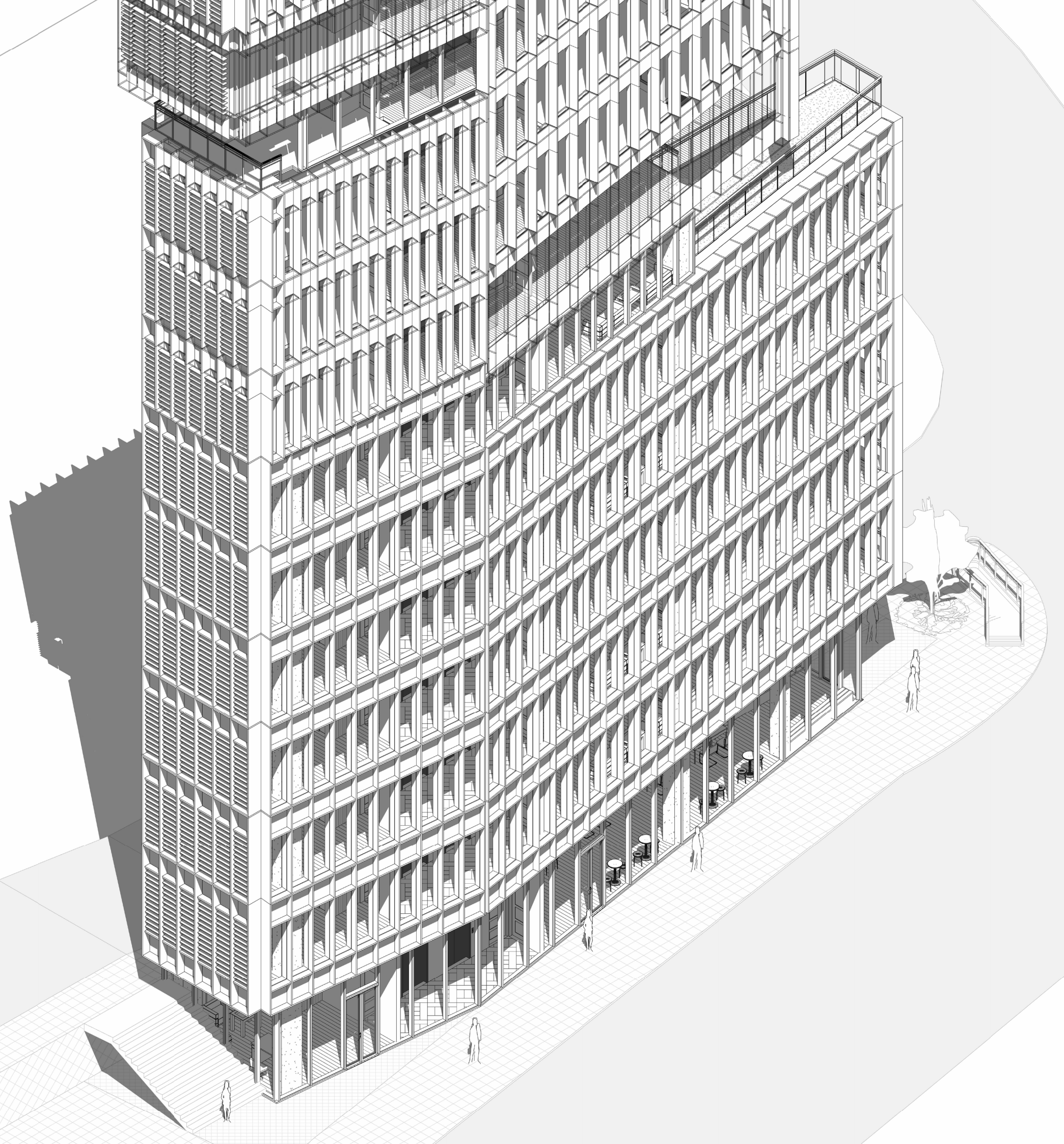
# CONTENT

INTRODUCTION

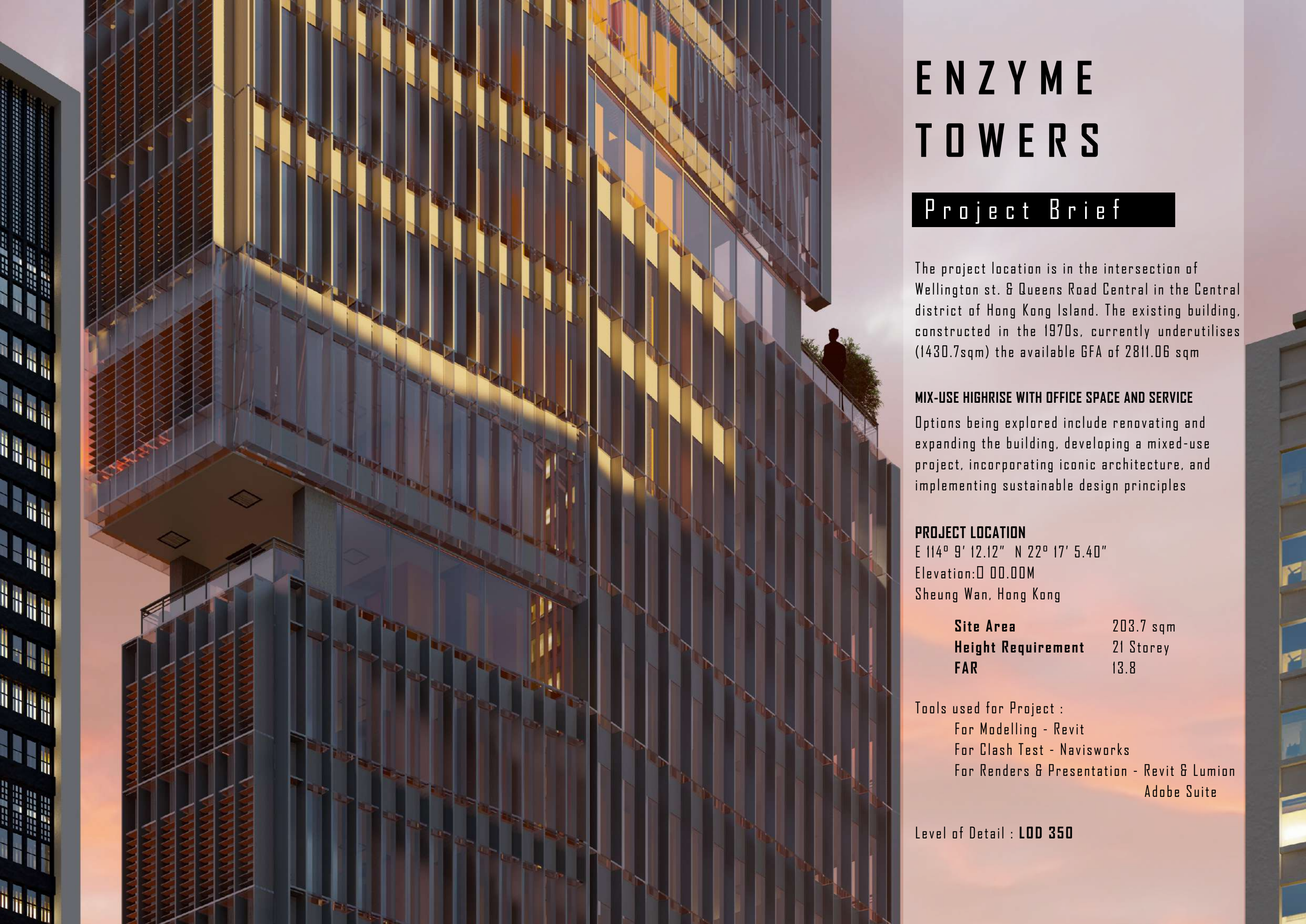
DESIGN

DRAWINGS

REPORTS







# ENZYME TOWERS

## Project Brief

The project location is in the intersection of Wellington st. & Queens Road Central in the Central district of Hong Kong Island. The existing building, constructed in the 1970s, currently underutilises (1430.7sqm) the available GFA of 2811.06 sqm

### MIX-USE HIGHRISE WITH OFFICE SPACE AND SERVICE

Options being explored include renovating and expanding the building, developing a mixed-use project, incorporating iconic architecture, and implementing sustainable design principles

### PROJECT LOCATION

E 114° 9' 12.12" N 22° 17' 5.40"

Elevation: 00.00M

Sheung Wan, Hong Kong

|                           |           |
|---------------------------|-----------|
| <b>Site Area</b>          | 203.7 sqm |
| <b>Height Requirement</b> | 21 Storey |
| <b>FAR</b>                | 13.8      |

Tools used for Project :

For Modelling - Revit

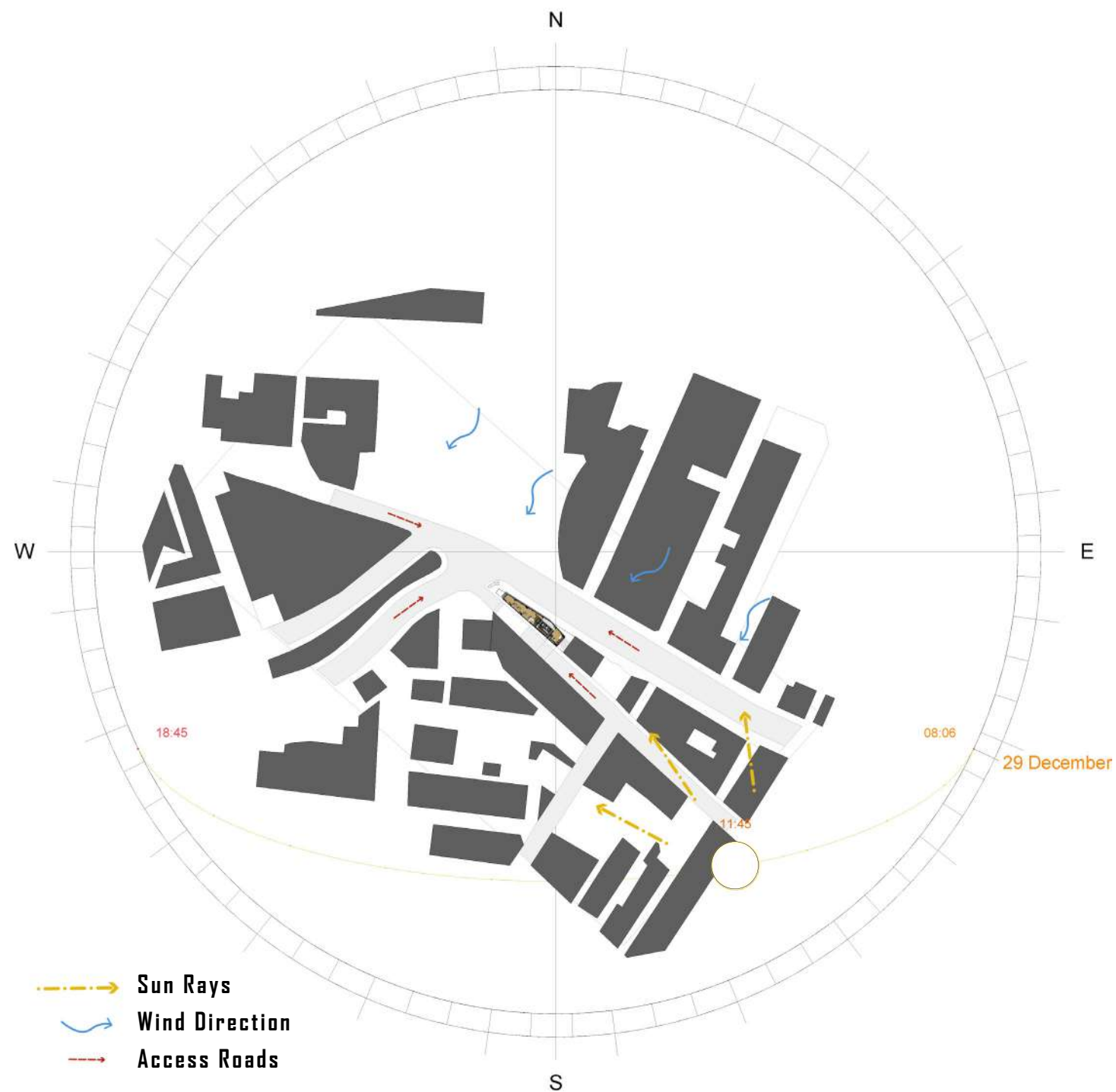
For Clash Test - Navisworks

For Renders & Presentation - Revit & Lumion  
Adobe Suite

Level of Detail : LOD 350



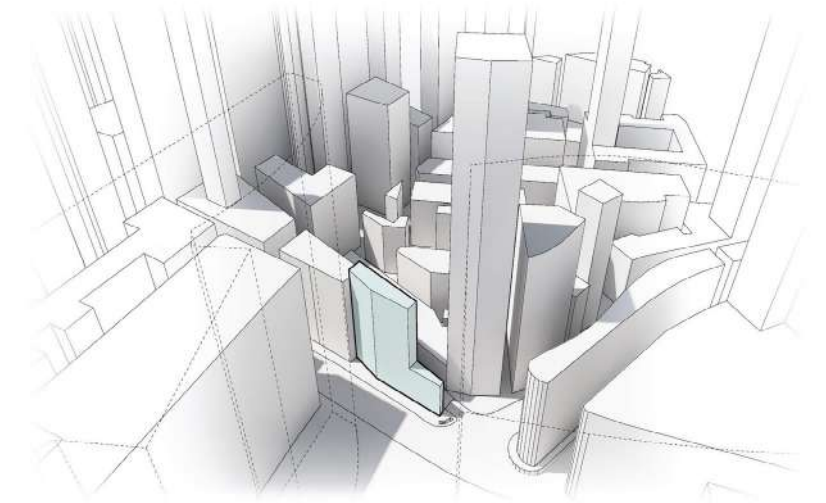
# SITE ANALYSIS



The project site's location at the intersection of Wellington St. and Queens Road Central is undeniably one of the most strategic and coveted spots in Hong Kong Island's central district. Understanding the sun and wind direction as well as access roads to a site is crucial for various reasons, including optimizing natural light and ventilation, ensuring efficient traffic flow, and overall enhancing the usability and sustainability of the development

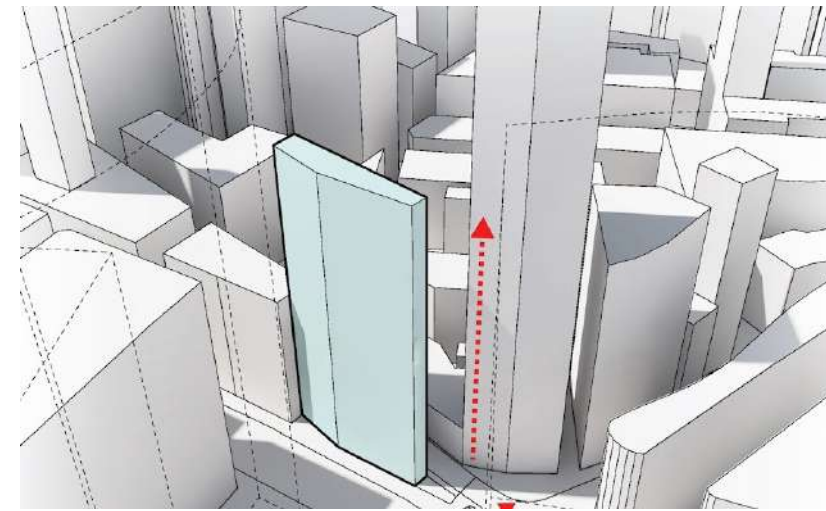
## Stage 0

Current building volume  
Understanding the limitations  
and the site and its built up



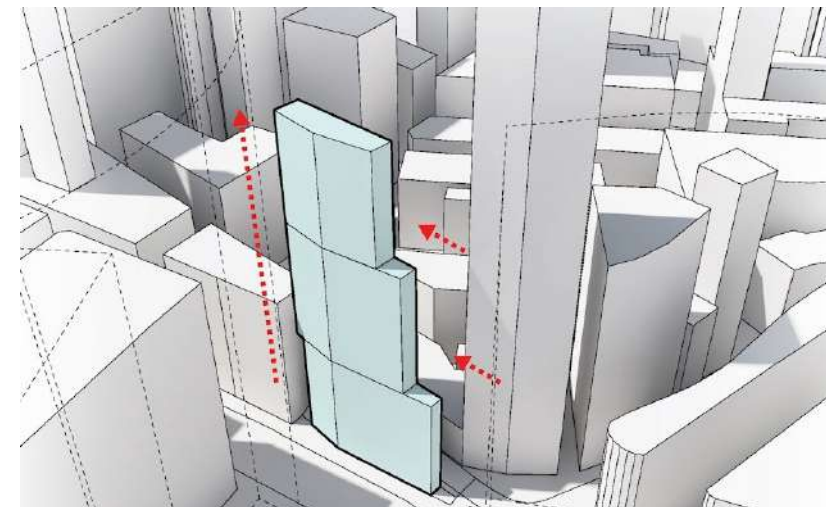
## Stage 1

Extrude to the maximum allowed  
number of stories which is 21  
and gives us the total height



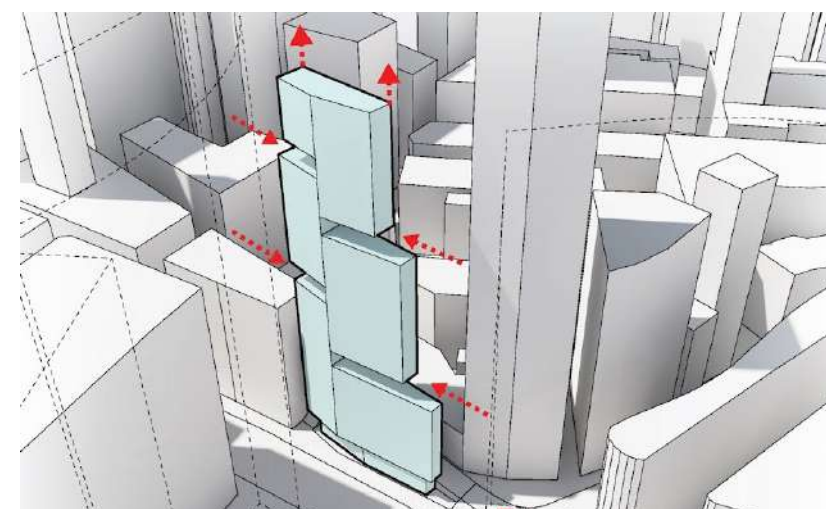
## Stage 2

Creating a series of set-backs  
allowed as per the local regulation  
of the addition of extra  
stories, in order to make up for the  
lost GFA



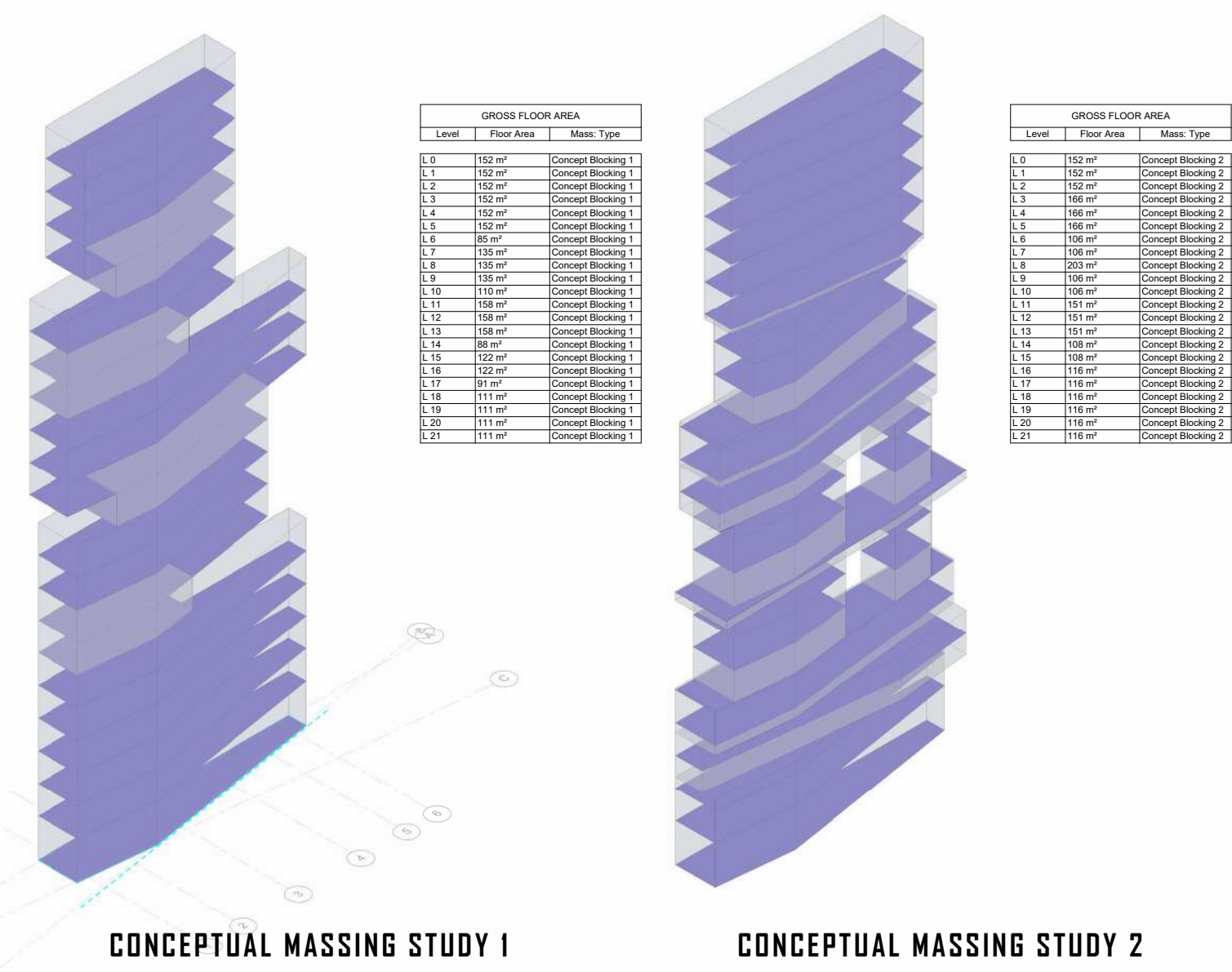
## Stage 3

The creation of sky gardens and  
public spaces, also allows the  
volume to grow higher recovering  
GFA, increasing the final number of  
stories





# MASSING



## Concept Design Study

In Revit, the “In-Place Mass” and “Design Options” tools are powerful features used for exploring conceptual massing volumes and refining design iterations.

Compare the GFA of each design option to identify which configuration yields the maximum allowed GFA per floor. This involves analyzing factors like floor area efficiency, circulation space, and utilization of available space

# MODEL BREAKDOWN

...

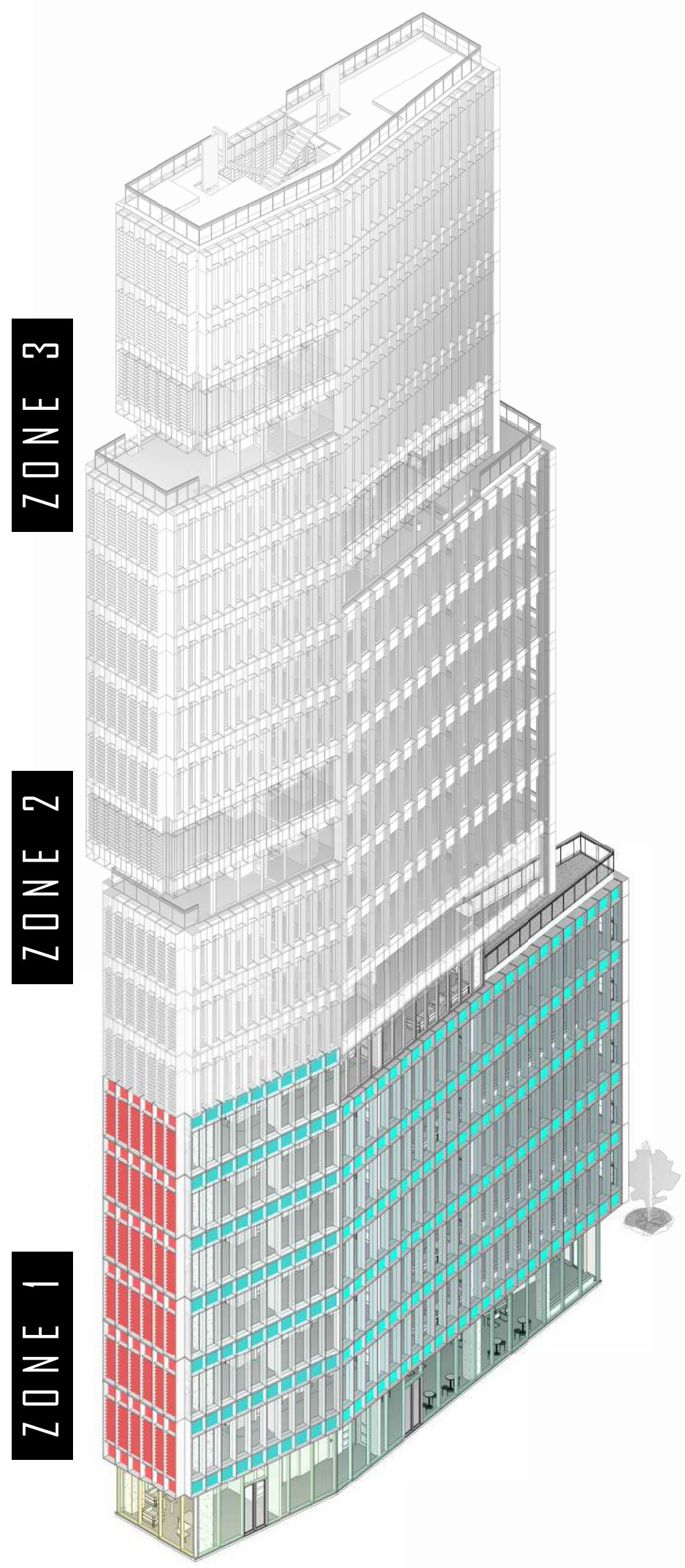
Multiple models for each discipline as per a portion (a group of levels) of the project, and the architecture model used for documentation The Architectural and Structural models will follow a separation by floors. Structural models should also follow construction joints.

Architectural models for the façade shall be done according in the same proportion as the division by groups (e.g.: two floors = 1 façade model)

- Architect 1 : Zone 1 = B1 to L6
- Architect 2 : Zone 2 = L7 to L14
- Architect 3 : Zone 3 = L15 to L21

The MEP models was provided for the coordination purposes.

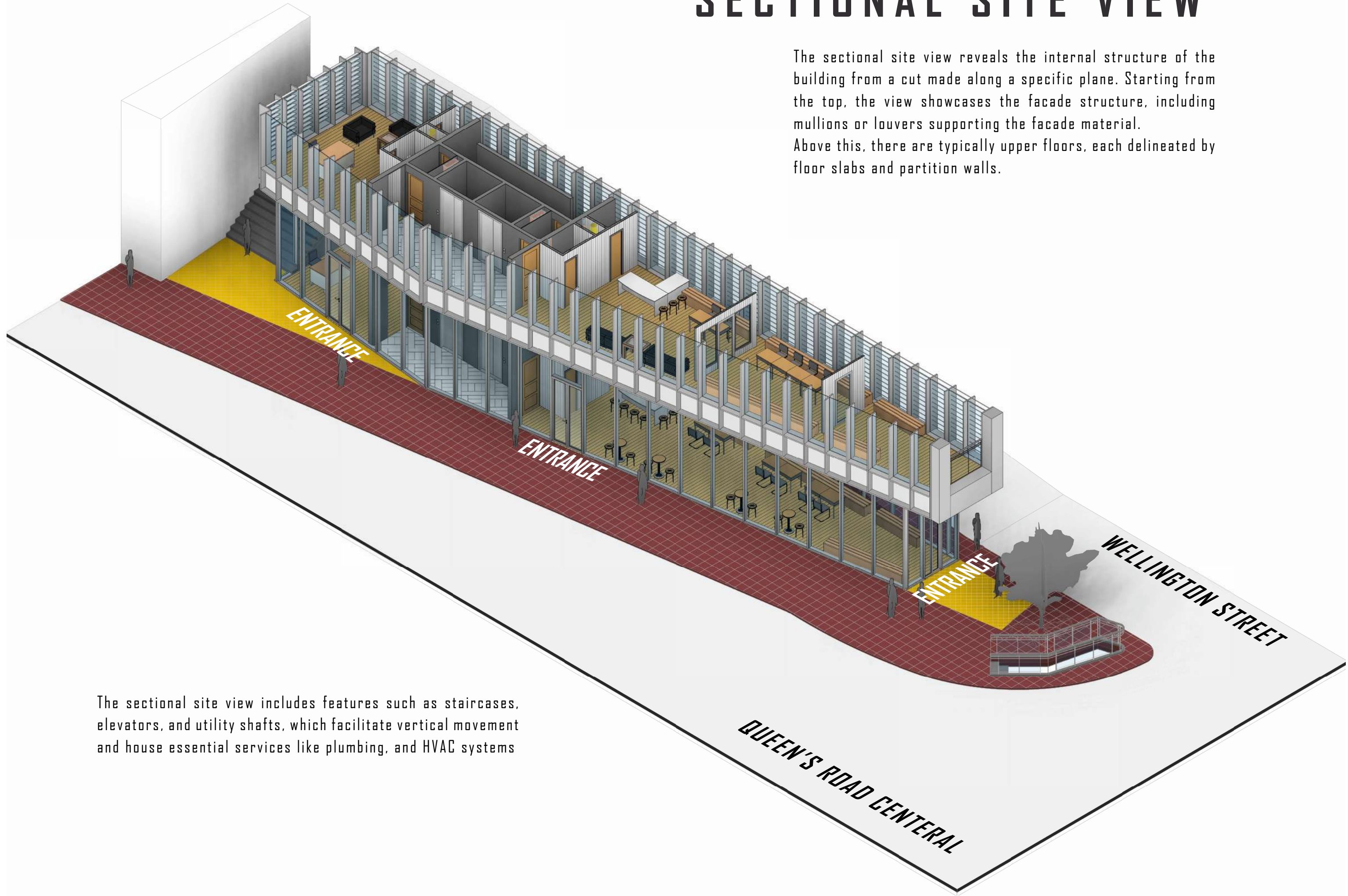
- The following AEC (UK) BIM protocol for Revit workflow was adopted:
- Create model file
  - Duplicate model file
  - Isolate model sub-divisions
  - Define and publish shared coordinates
  - Linked files will be visible





# SECTIONAL SITE VIEW

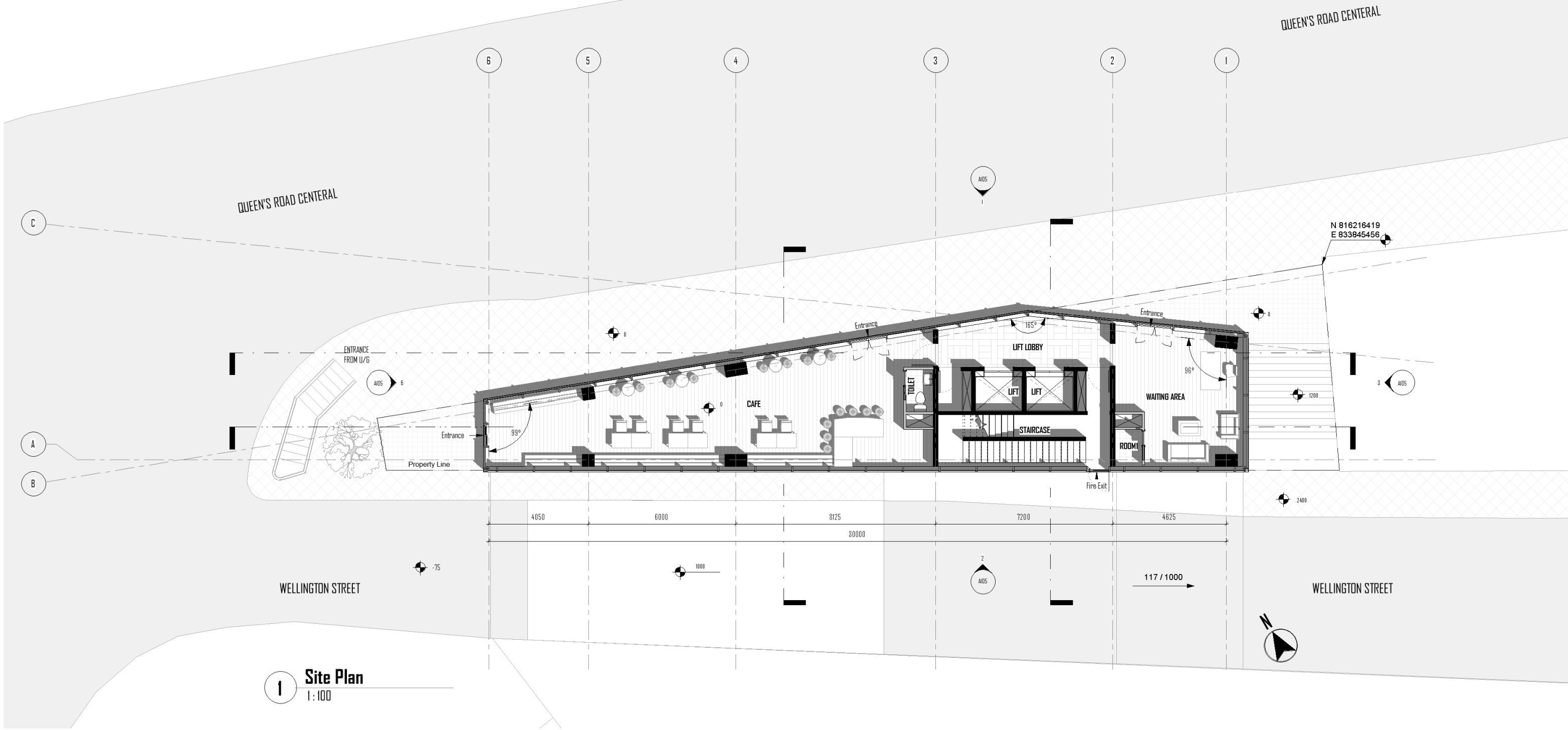
The sectional site view reveals the internal structure of the building from a cut made along a specific plane. Starting from the top, the view showcases the facade structure, including mullions or louvers supporting the facade material. Above this, there are typically upper floors, each delineated by floor slabs and partition walls.



The sectional site view includes features such as staircases, elevators, and utility shafts, which facilitate vertical movement and house essential services like plumbing, and HVAC systems



# SITE PLAN



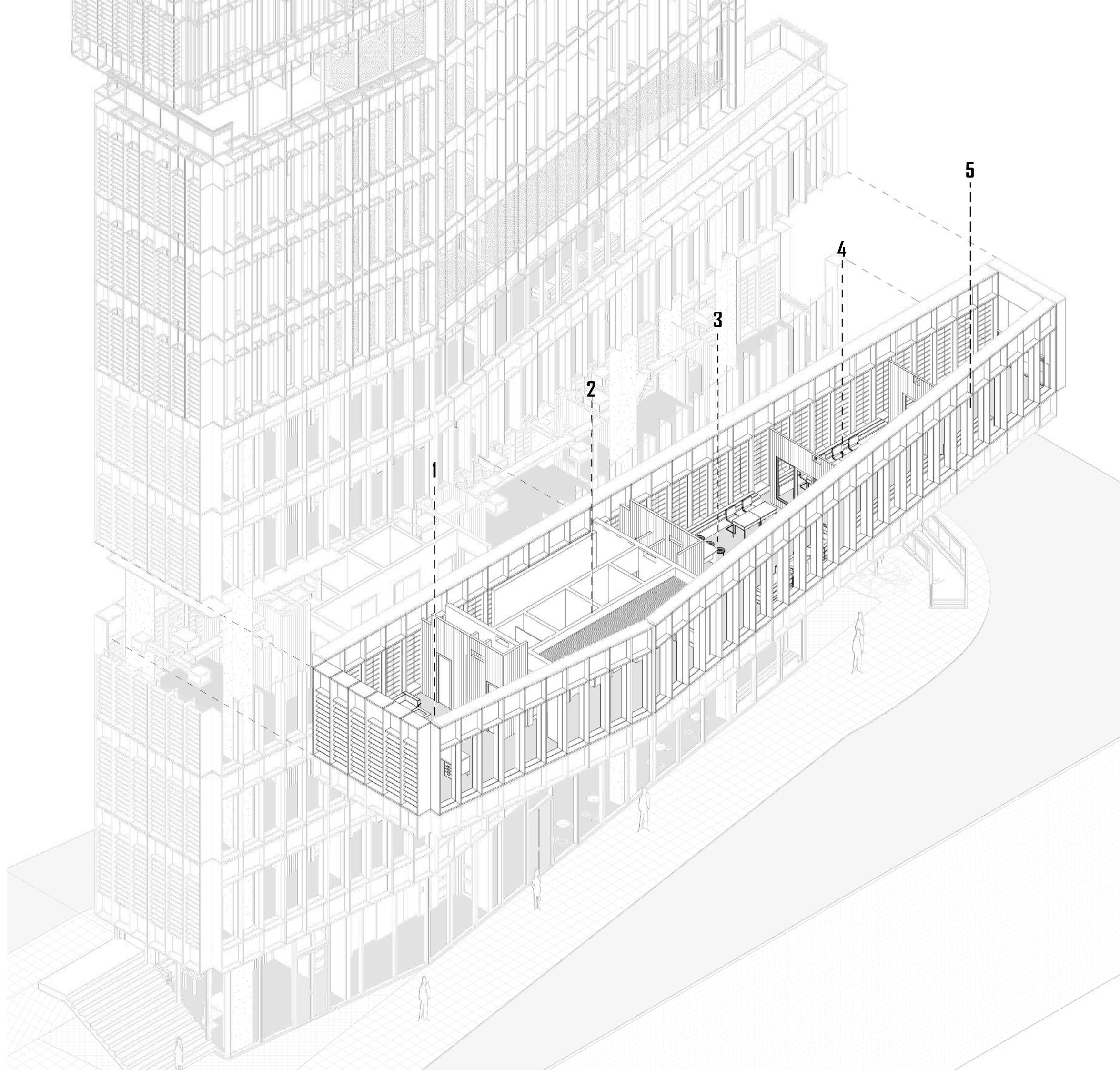
Detailing the site plan of the Sheung Wan Tower is essential to understand its layout, functionality, and overall design integration within the urban landscape

The Sheung Wan Tower is strategically located in the heart of Hong Kong's bustling business district, amidst a dynamic blend of commercial, residential, recreational spaces. Its site covers an approximate area of 203 square meters. It serves as a model for contemporary high-rise developments in dense urban environments like Hong Kong.



# FLOOR PLANS

- 1 OFFICE AREA 1
- 2 LIFT LOBBY & BUILDING CORE
- 3 OFFICE AREA 2
- 4 OFFICE AREA 3
- 5 OFFICE AREA 4





# TYPICAL FLOOR PLAN



Level 1 is the Typical floor plan, Having the core in the center of the building and has two major sellable spaces with diffrent areas and utilities.

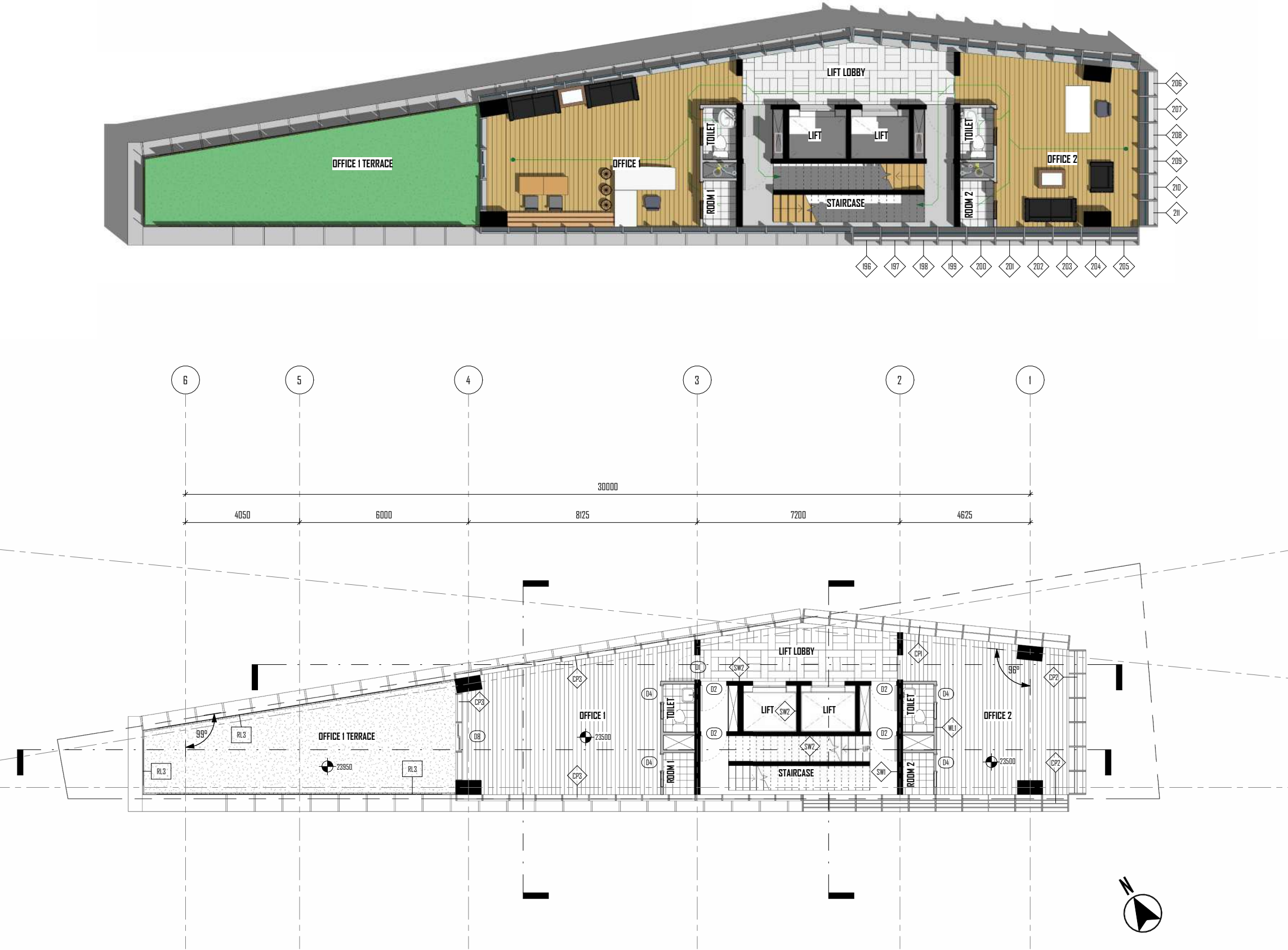
This floor also has rotateable louvres to limit the sun exposure on its SW and SE facade



# ATYPICAL FLOOR PLAN

Level 6 is again a atypical plan for zone I consisting two office space an attached office terrace with office space I.This open terrace also acts as the green space for the building.

In addition,it has attached washrooms with utilities room



| Room Schedule L6 |                   |                    |
|------------------|-------------------|--------------------|
| Name             | Level             | Area               |
| LIFT             | L 6 Atypical Plan | 3 m <sup>2</sup>   |
| LIFT             | L 6 Atypical Plan | 3 m <sup>2</sup>   |
| LIFT LOBBY       | L 6 Atypical Plan | 13 m <sup>2</sup>  |
| OFFICE 1         | L 6 Atypical Plan | 35 m <sup>2</sup>  |
| OFFICE 1 TERRACE | L 6 Atypical Plan | 36 m <sup>2</sup>  |
| OFFICE 2         | L 6 Atypical Plan | 27 m <sup>2</sup>  |
| ROOM 1           | L 6 Atypical Plan | 2 m <sup>2</sup>   |
| ROOM 2           | L 6 Atypical Plan | 2 m <sup>2</sup>   |
| STAIRCASE        | L 6 Atypical Plan | 14 m <sup>2</sup>  |
| TOILET           | L 6 Atypical Plan | 2 m <sup>2</sup>   |
| TOILET           | L 6 Atypical Plan | 2 m <sup>2</sup>   |
| Grand total: 11  |                   | 137 m <sup>2</sup> |

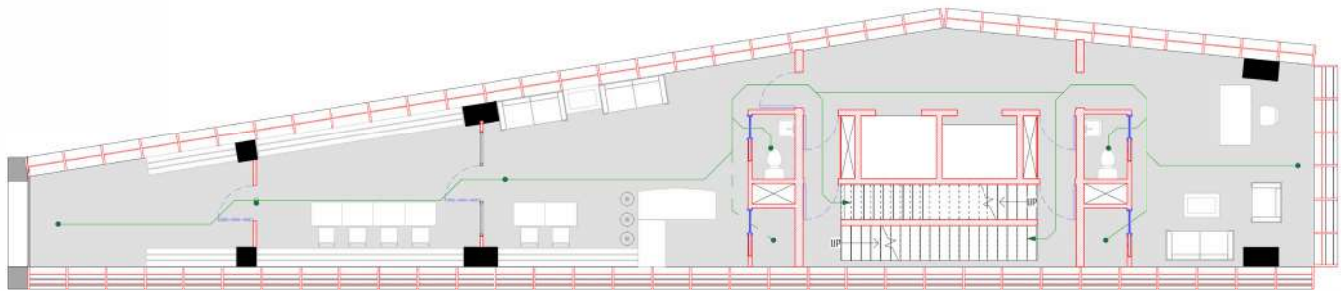
| Area Schedule by Level 6 |                   |                    |
|--------------------------|-------------------|--------------------|
| Level                    | Name              | Area               |
| L 6 Atypical Plan        | NON SELLABLE AREA | 43 m <sup>2</sup>  |
| L 6 Atypical Plan        | NON SELLABLE AREA | 37 m <sup>2</sup>  |
| L 6 Atypical Plan        | SELLABLE AREA     | 42 m <sup>2</sup>  |
| L 6 Atypical Plan        | SELLABLE AREA     | 31 m <sup>2</sup>  |
| Total Area: 4            |                   | 153 m <sup>2</sup> |

| Door Schedule by Level 6 |             |       |           |       |
|--------------------------|-------------|-------|-----------|-------|
| Door Name                | Head Height | Width | FFR       | Count |
| D1                       | 2400        | 900   | 120.0 min | 1     |
| D2                       | 2100        | 900   | 120.0 min | 4     |
| D4                       | 2062        | 586   | 120.0 min | 4     |
| D8                       |             | 989   | 120.0 min | 1     |
| Total No of Doors: 10    |             |       |           | 10    |

| Panel Schedule L6       |                   |                |       |       |
|-------------------------|-------------------|----------------|-------|-------|
| Type                    | Facade Identifier | Rotation Angle | Level | Count |
| DNX_CAP_ARC_CUR_GLZ_LOV | SE                | 15.00°         | L6    | 6     |
| DNX_CAP_ARC_CUR_GLZ_LOV | SW                | 45.00°         | L6    | 10    |
| Total No of Panels      |                   |                |       | 16    |



# FIRE, AREA, COORDINATION & CEILING PLANS

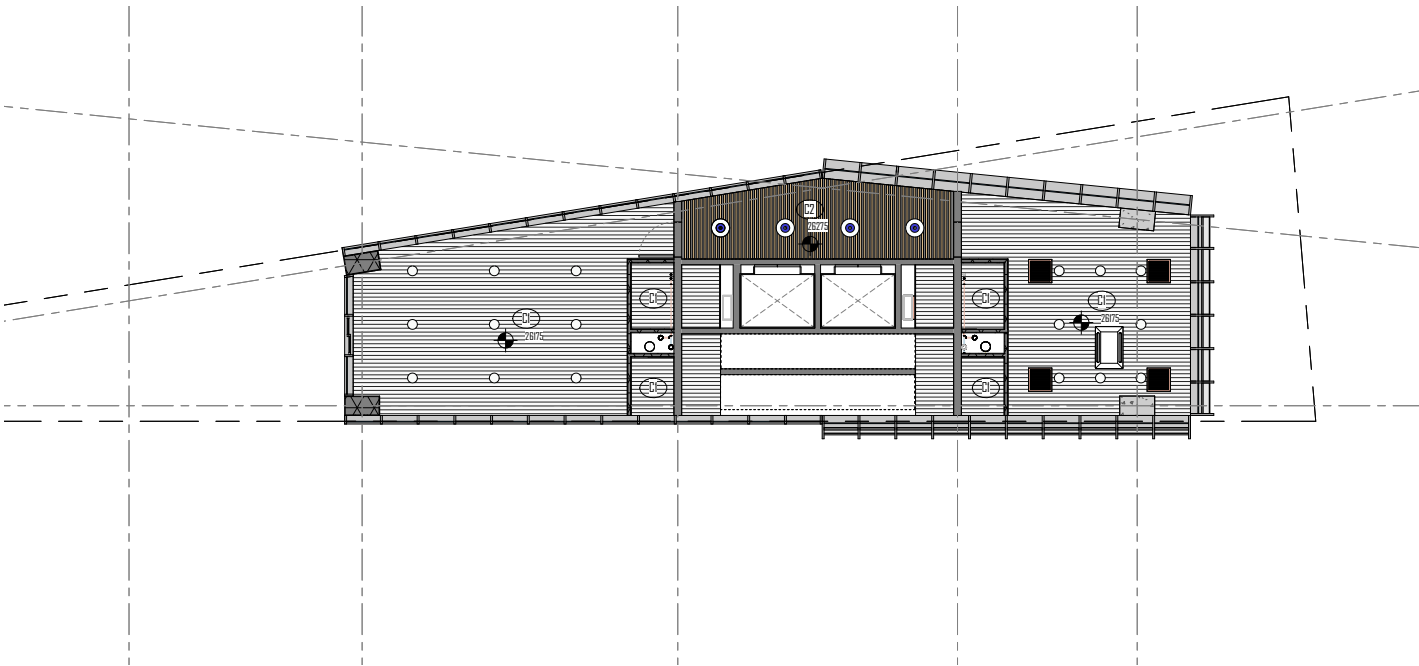
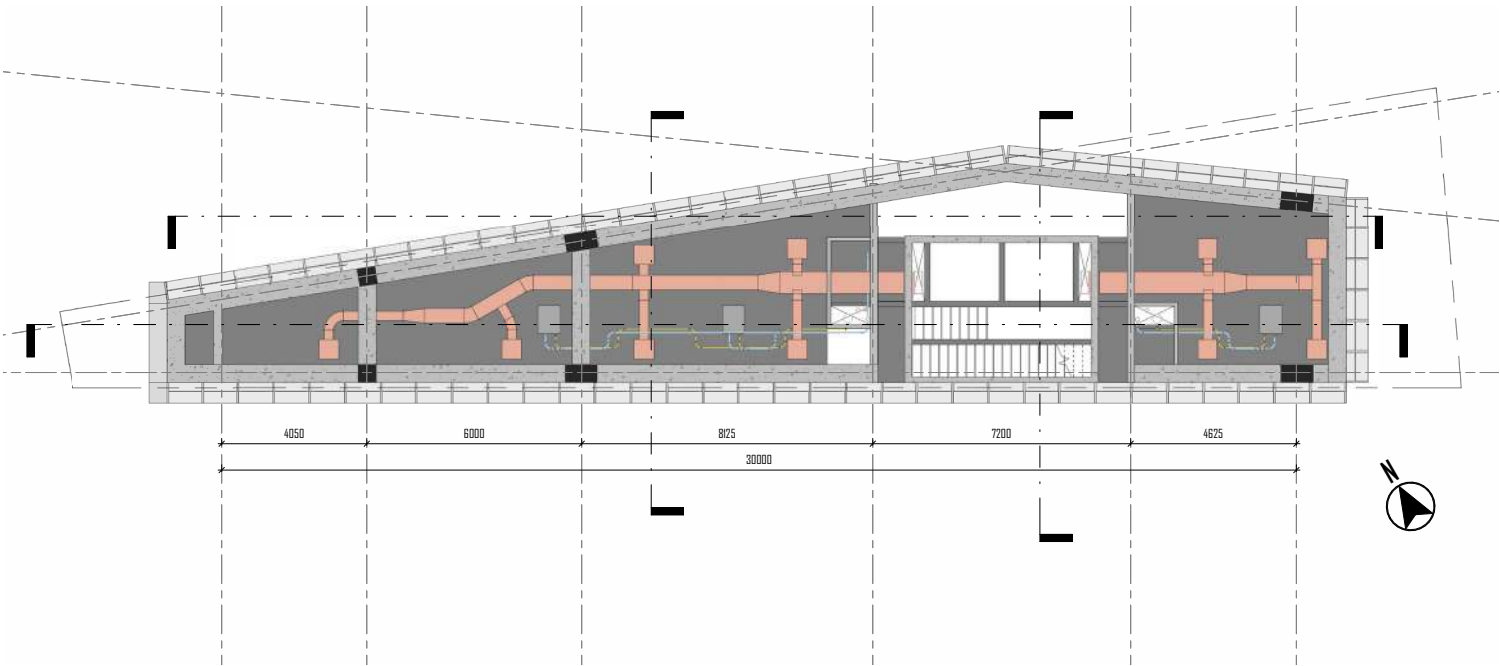
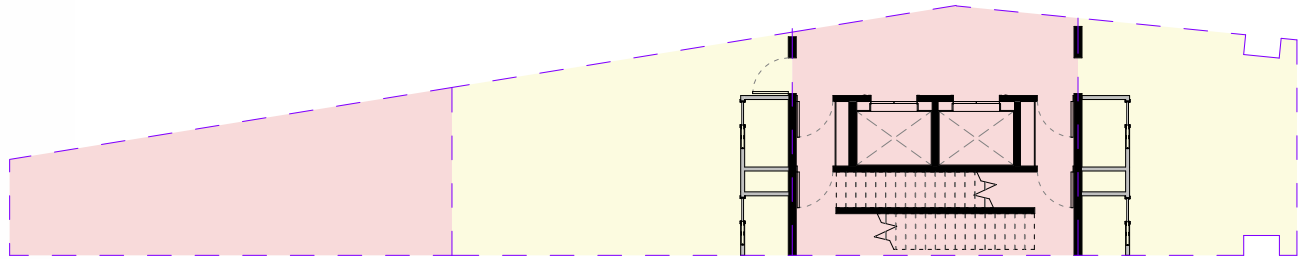
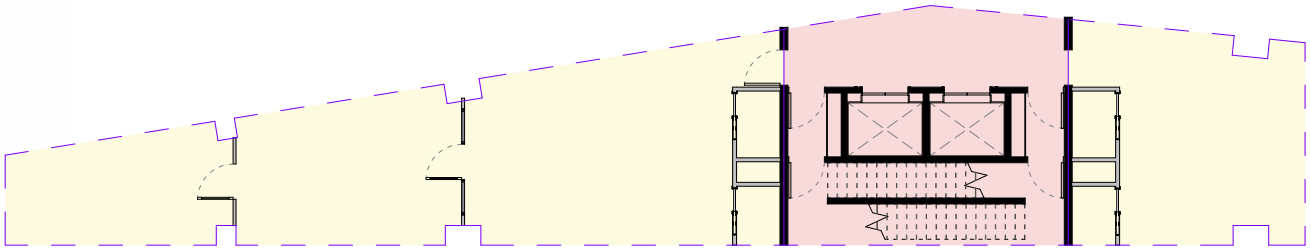


| Surface: | Surface: | Name             | Type  | Fire Rating |
|----------|----------|------------------|-------|-------------|
| —        | —        | Fire Safety_Wall | Walls | 60 mins     |
| - - -    | —        | Fire Safety_Door | Doors | 120 mins    |
| —        | —        | Path of Travel   |       |             |



AREA SCHEME

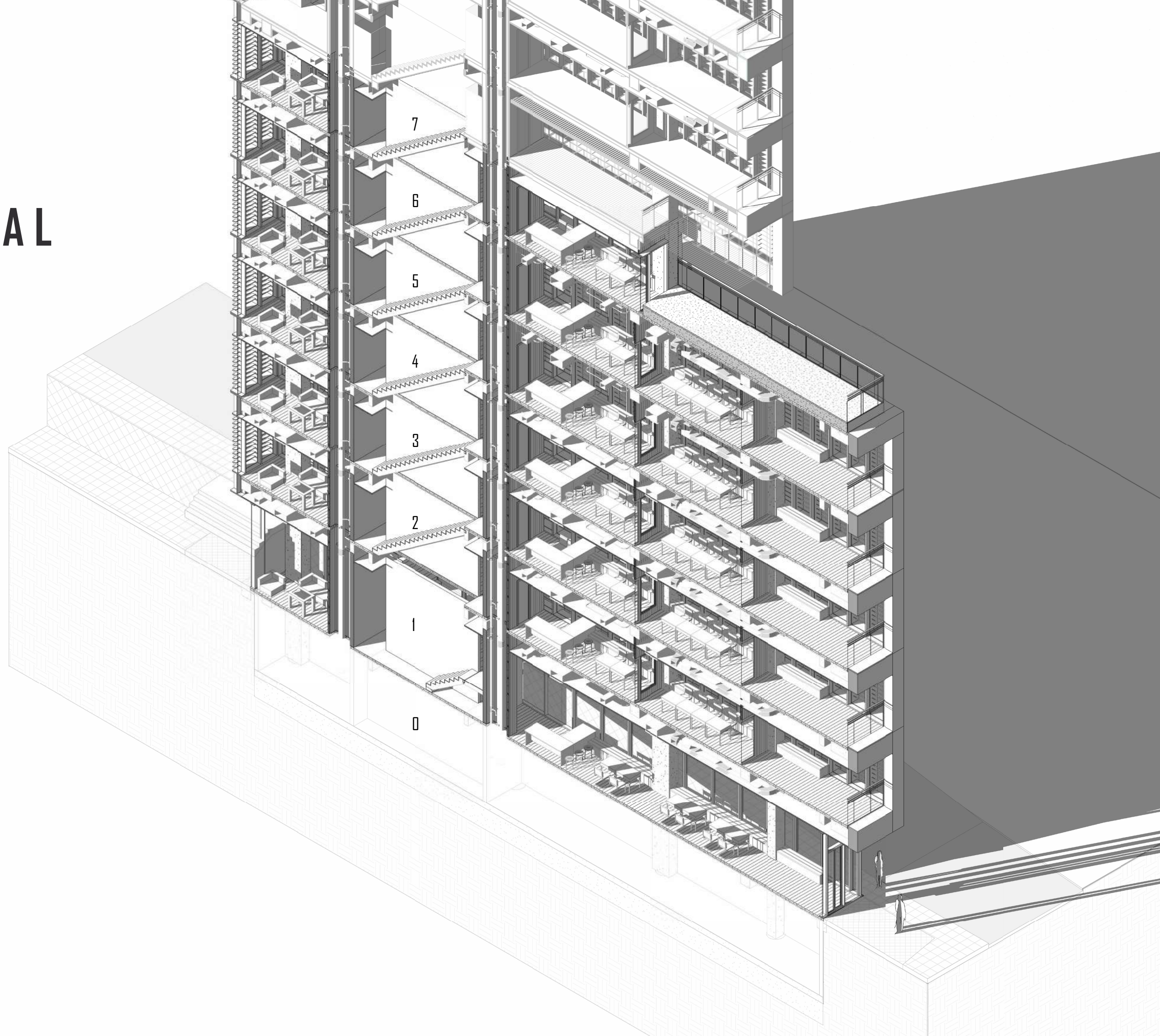
- NON SELLABLE AREA
- SELLABLE AREA



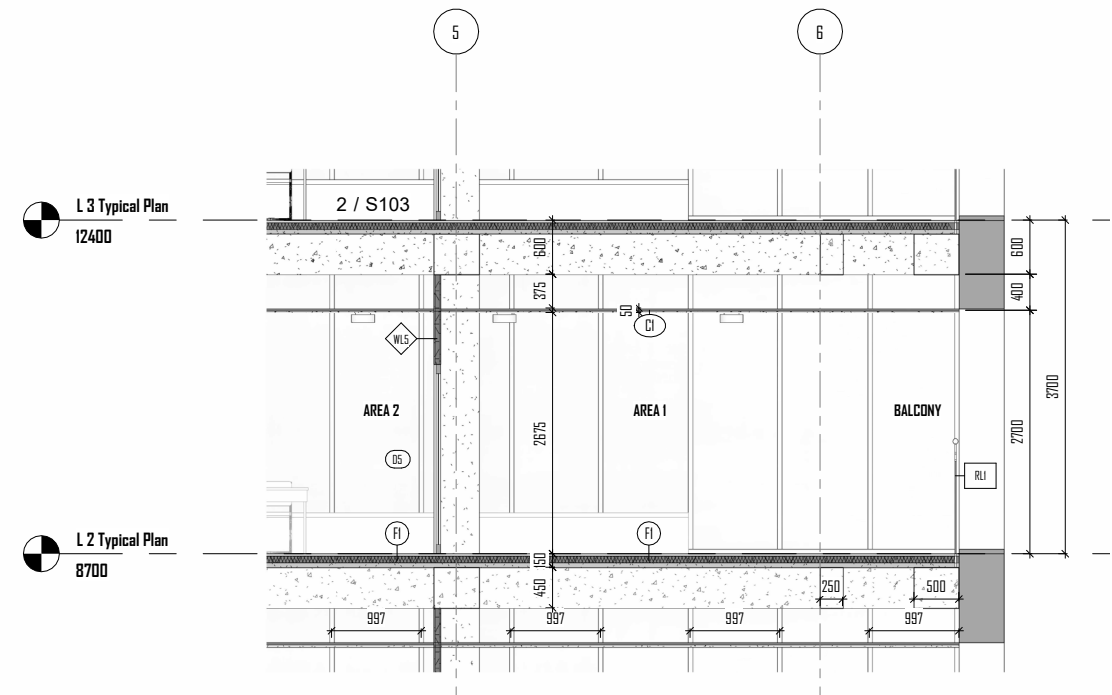
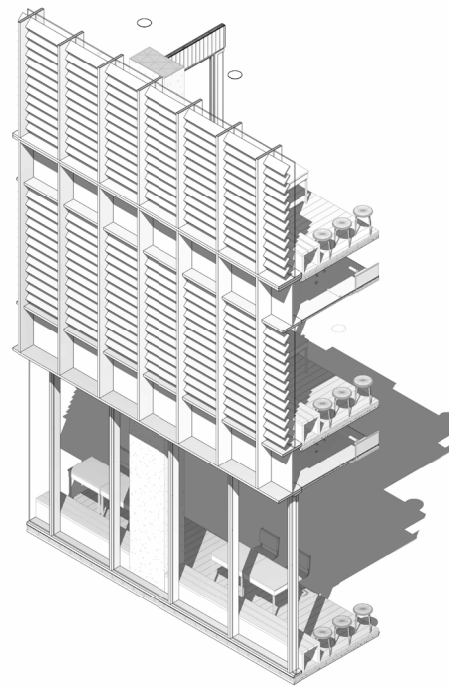
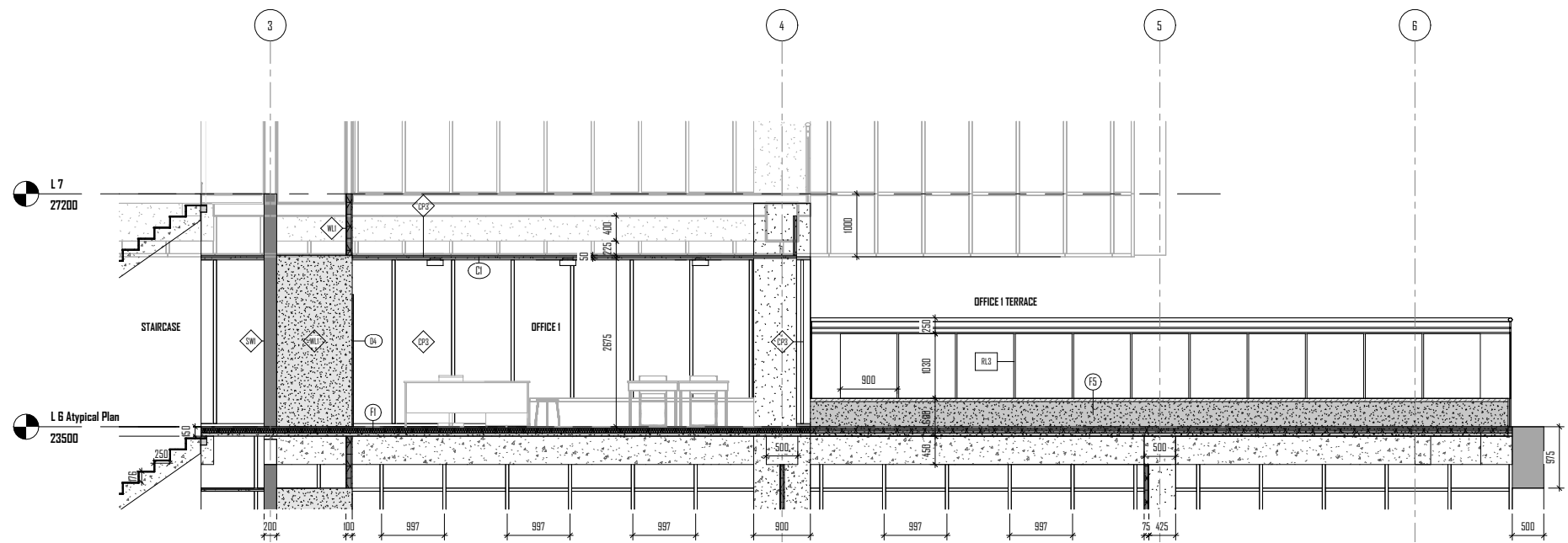
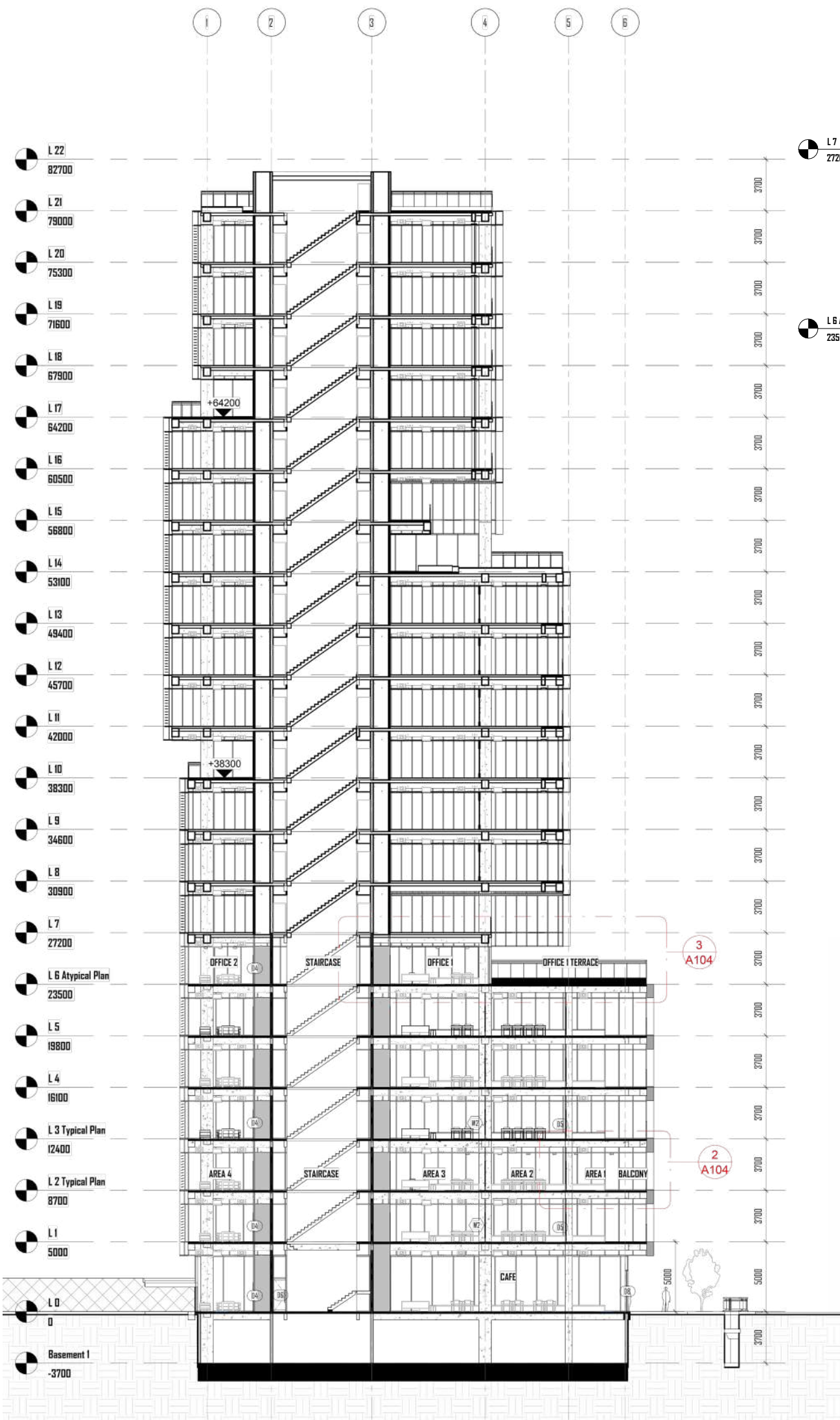


# SECTIONAL DETAILS

- 0 BASEMENT 1
- 1 GROUND FLOOR
- 2 LEVEL 1
- 3 LEVEL 2
- 4 LEVEL 3
- 5 LEVEL 4
- 6 LEVEL 5
- 7 LEVEL 6

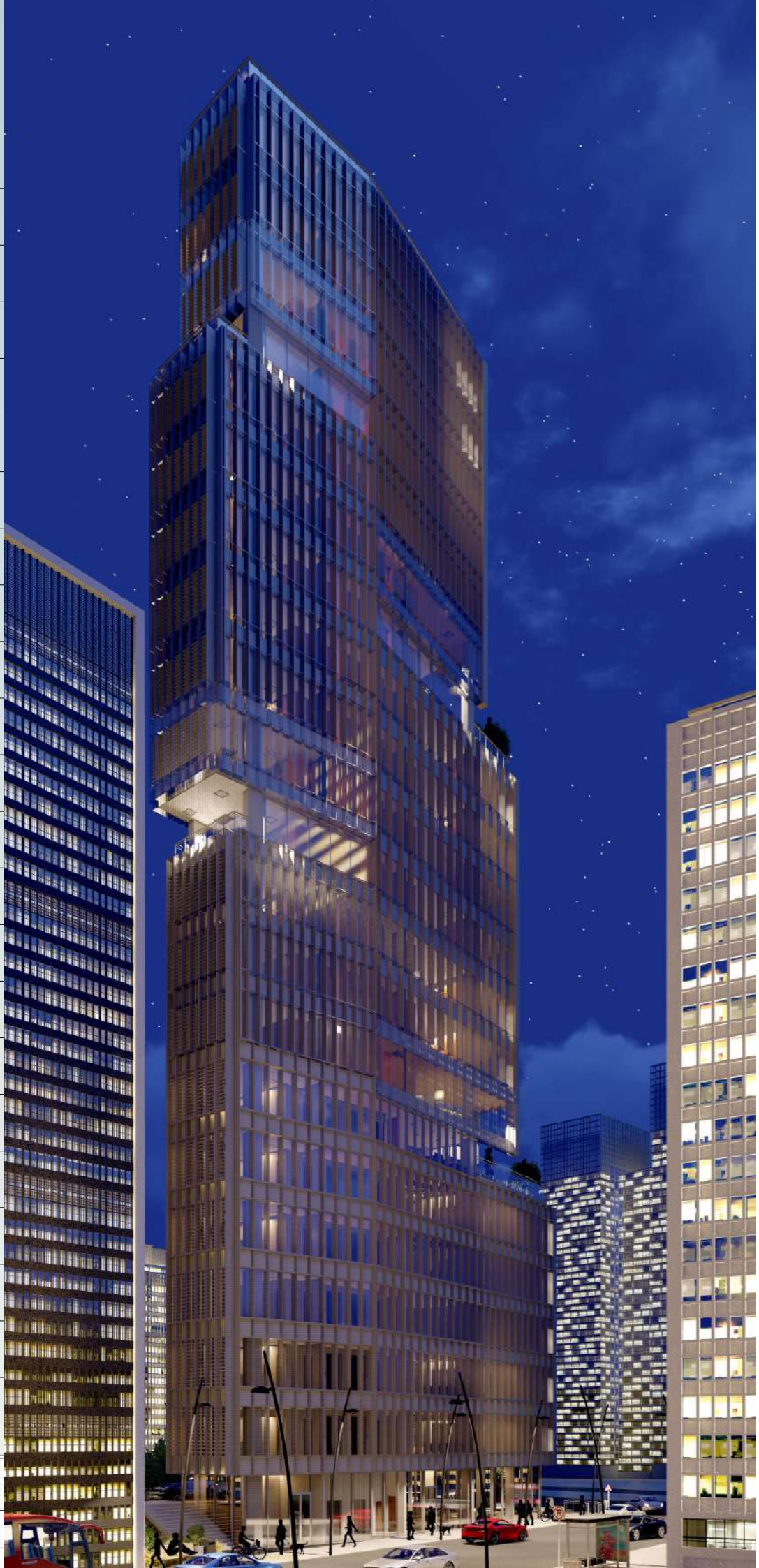
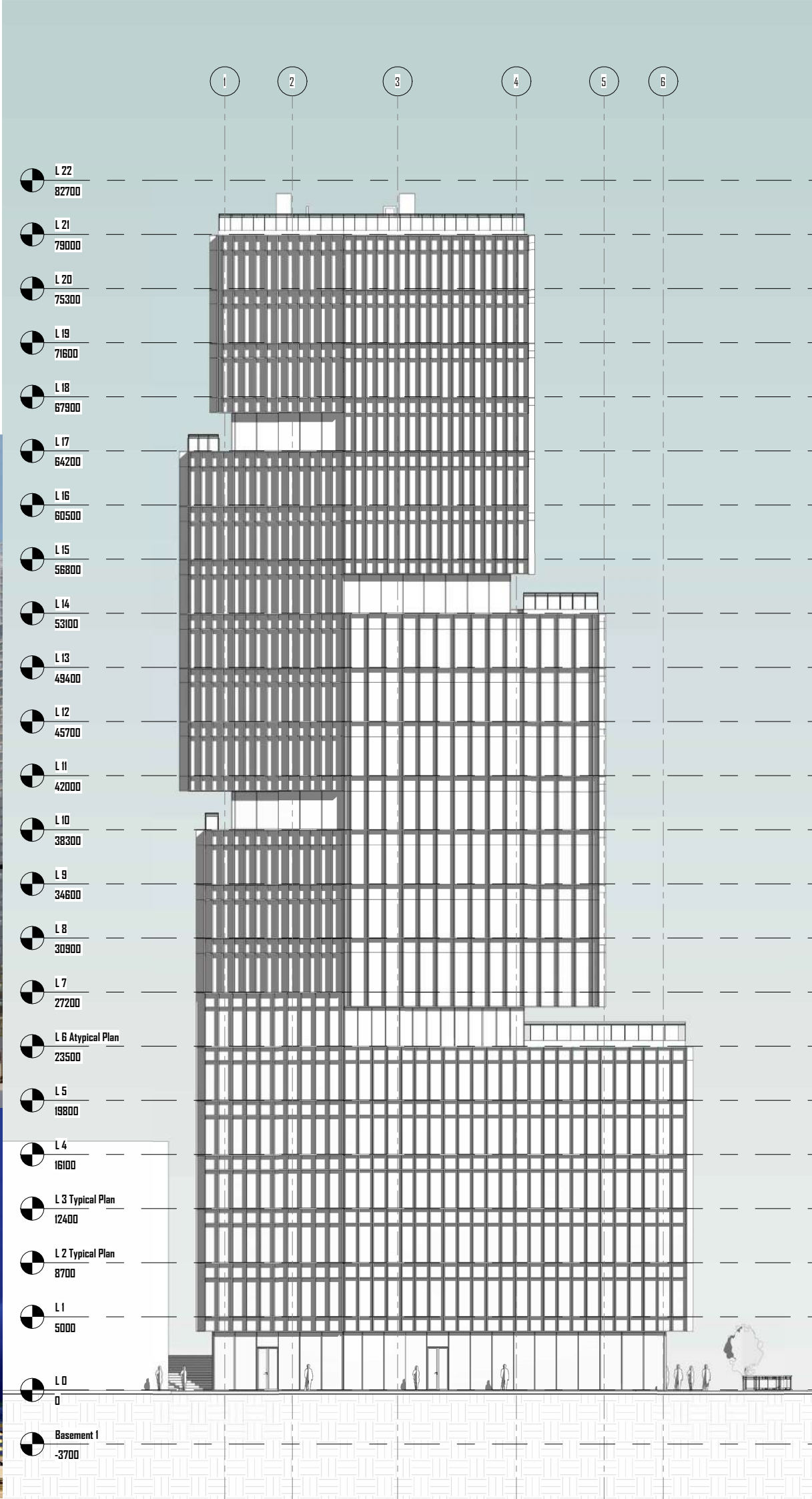




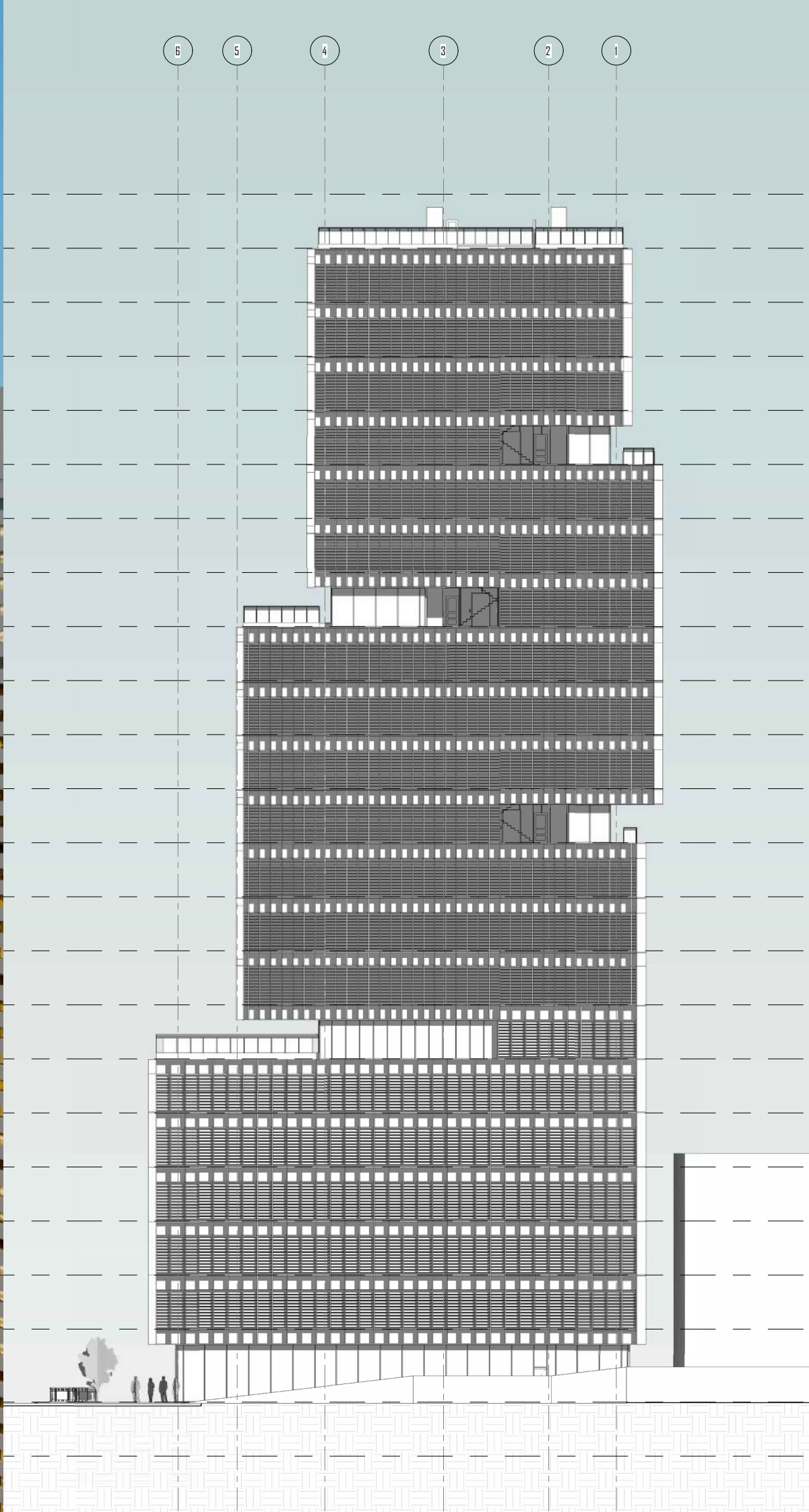




# NORTH ELEVATIONS

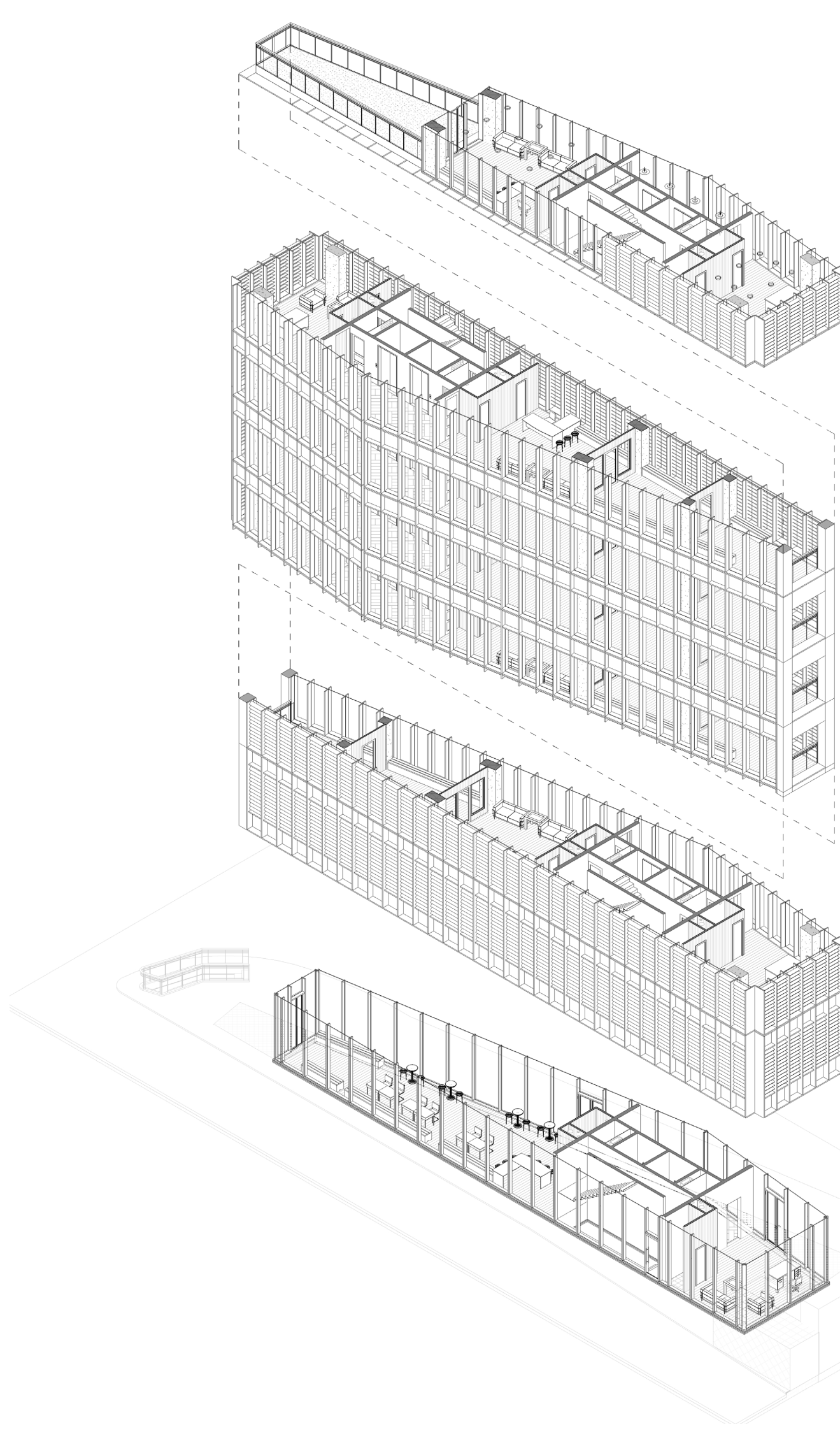
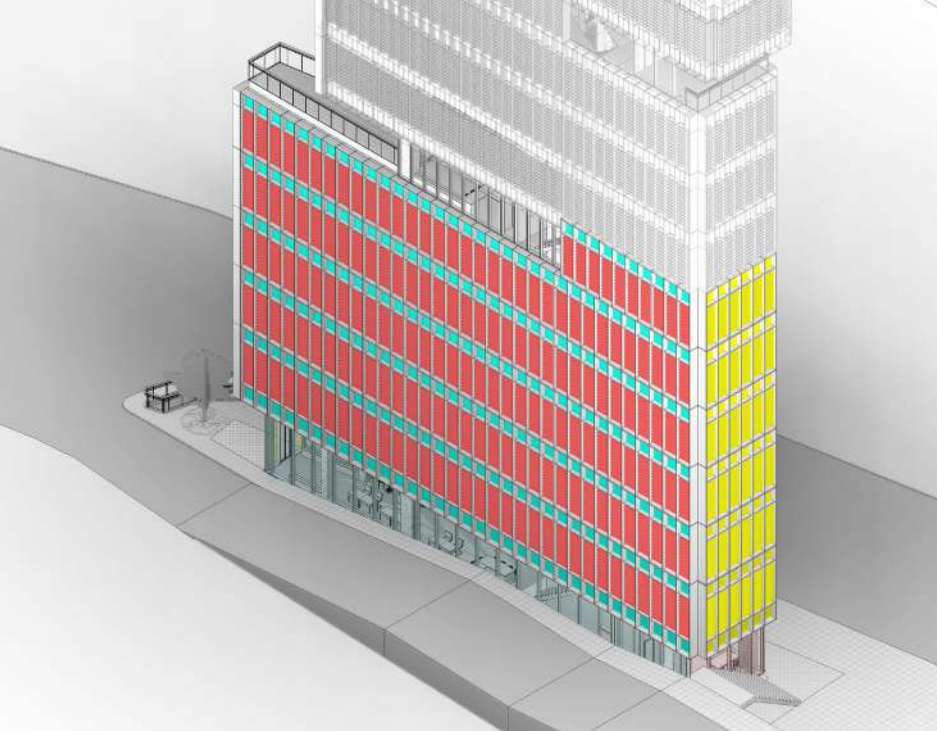






**SOUTH  
ELEVATIONS**





Level 6 is again a atypical plan for zone 1 consisting two office space an attached office terrace with office space 1.This open terrace also acts as the green space for the building. In addition has attached washrooms with utilities room

Level 2-5 is the Typical floor plan same as level 1 but with small panels , Having the core in the center of the building and has tw major sellable spaces with different areas and utilities. This floor also has rotateable louvres to limit the sun exposure on its SW ; SE facade

Level 1 is the Typical floor plan, Having the core in the center of building and has two major sellable spaces with diffrent areas a utilities. This floor also has rotateable louvres to limit the sun exposure on its SW and SE facade

Level 0 is the ground plan and demonstrates compliance with regulations by providing generous open space around the buildi The ground floor is designed as both an entrance lobby for the building and a potential public area.



# INTRODUCTION TO CLASH RESOLUTION

Clash resolution involves the systematic process of addressing and resolving conflicts or clashes identified during the clash detection

## NAVISWORKS CLASH DETECTION MATRIX

| CLASH DETECTION MATRIX |              | ARC          | STR     | MEP | SIT  |
|------------------------|--------------|--------------|---------|-----|------|
|                        |              | WALL & FLOOR | STRUCT. | MEP | SITE |
| ARC                    | WALL & FLOOR | N/A          | N/A     | N/A | N/A  |
| STR                    | STRUCT.      | T01          | N/A     | N/A | N/A  |
| MEP                    | MEP          | T02          | T03     | T04 | N/A  |
| SIT                    | SITE         | N/A          | N/A     | N/A | N/A  |

N/A

Low Priority

High Priority

## SEARCH SETS

| DISCIPLINE         | NAME OF THE SET | DESCRIPTION                                                          |
|--------------------|-----------------|----------------------------------------------------------------------|
| ARC-All            | [ARC]-Set01     | Architectural walls (interior, exterior) and foors                   |
| ARC-Walls          | [ARC]-Set02     | Architectural walls (interior, exterior)                             |
| ARC-Floors         | [ARC]-Set03     | Architectural Floors                                                 |
| STR-All            | [STR]-Set01     | Structural columns and structural framing (beams) and other elements |
| STR-Primary Str.   | [STR]-Set02     | Structural columns and structural framing (beams)                    |
| STR-Secondary Str. | [STR]-Set03     | Other (Structural wall, floors, stair, ramps etc.)                   |
| MEP-All            | [MEP]-Set01     | MEP elements                                                         |

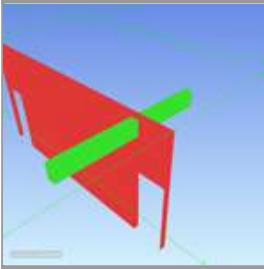
## CLASH TESTS

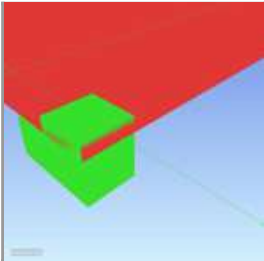
| Test No | Selection A | Selection B | Test Name                   | Priority | Rules | Select              |      |      |
|---------|-------------|-------------|-----------------------------|----------|-------|---------------------|------|------|
|         |             |             |                             |          |       | Type                | Link | Tol. |
| T01     | ARC-All     | STR-All     | T01_[ARC]-Set01-[STR]-Set01 | Low      |       | Hard                | None | 5 cm |
| T02     | ARC-All     | MEP-All     | T02_[ARC]-Set01-[MEP]-Set01 | Low      |       | Hard                | None | 5 cm |
| T03     | STR-All     | MEP-All     | T03_[STR]-Set01-[MEP]-Set01 | High     |       | Hard                | None | 5 cm |
| T04     | MEP-All     | MEP-All     | T04_[MEP]-Set01-[MEP]-Set01 | High     |       | Hard (Conservative) | None | 2 cm |
| NA      | SIT-All     | SIT-All     | N/A                         | NA       |       | Hard                | None | 5 cm |

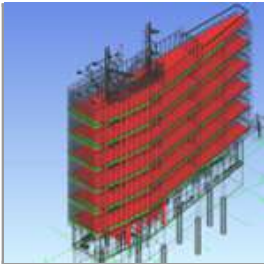


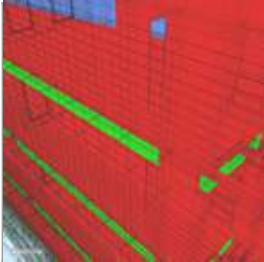
# INITIAL CLASH DETECTION

| T01_[ARC]-Set01-[STR]-Set01 | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type | Status |
|-----------------------------|-----------|---------|-----|--------|----------|----------|----------|------|--------|
|                             | 0.050m    | 76      | 0   | 76     | 0        | 0        | 0        | Hard | OK     |

|                                                                                  |        |        |                        |      |                 |       |                              |                    |              |       |                    |                              |       |
|----------------------------------------------------------------------------------|--------|--------|------------------------|------|-----------------|-------|------------------------------|--------------------|--------------|-------|--------------------|------------------------------|-------|
|  | Clash4 | Active | A-2 : L 2 Typical Plan | Hard | 2024/2/18 23:48 | Vivek | x:-9.520, y:-6.100, z:16.316 | Element ID: 874866 | Default Wall | Solid | Element ID: 717913 | Concrete, Cast-in-Place gray | Solid |
|----------------------------------------------------------------------------------|--------|--------|------------------------|------|-----------------|-------|------------------------------|--------------------|--------------|-------|--------------------|------------------------------|-------|

|                                                                                  |         |        |                        |      |                 |       |                              |                     |                              |       |                    |                              |       |
|----------------------------------------------------------------------------------|---------|--------|------------------------|------|-----------------|-------|------------------------------|---------------------|------------------------------|-------|--------------------|------------------------------|-------|
|  | Clash41 | Active | A-1 : L 3 Typical Plan | Hard | 2024/2/18 23:48 | Vivek | x:-2.528, y:-7.699, z:20.466 | Element ID: 1055159 | Concrete, Sand/Cement Screed | Solid | Element ID: 716228 | Concrete, Cast-in-Place gray | Solid |
|----------------------------------------------------------------------------------|---------|--------|------------------------|------|-----------------|-------|------------------------------|---------------------|------------------------------|-------|--------------------|------------------------------|-------|

|                                                                                   |         |        |           |      |                 |       |                             |                    |                              |       |                    |                              |       |
|-----------------------------------------------------------------------------------|---------|--------|-----------|------|-----------------|-------|-----------------------------|--------------------|------------------------------|-------|--------------------|------------------------------|-------|
|  | Clash52 | Active | A-1 : L 0 | Hard | 2024/2/18 23:48 | Vivek | x:-2.528, y:-7.699, z:9.366 | Element ID: 750575 | Concrete, Sand/Cement Screed | Solid | Element ID: 714566 | Concrete, Cast-in-Place gray | Solid |
|-----------------------------------------------------------------------------------|---------|--------|-----------|------|-----------------|-------|-----------------------------|--------------------|------------------------------|-------|--------------------|------------------------------|-------|

|                                                                                    |         |        |           |      |                 |       |                             |                    |                              |       |                    |                              |       |
|------------------------------------------------------------------------------------|---------|--------|-----------|------|-----------------|-------|-----------------------------|--------------------|------------------------------|-------|--------------------|------------------------------|-------|
|  | Clash17 | Active | A-3 : L 2 | Hard | 2024/2/18 23:48 | Vivek | x:-2.528, y:-7.699, z:9.366 | Element ID: 716240 | Concrete, Sand/Cement Screed | Solid | Element ID: 920830 | Concrete, Cast-in-Place gray | Solid |
|------------------------------------------------------------------------------------|---------|--------|-----------|------|-----------------|-------|-----------------------------|--------------------|------------------------------|-------|--------------------|------------------------------|-------|

| T03_[STR]-Set01-[MEP]-Set01 | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type | Status |
|-----------------------------|-----------|---------|-----|--------|----------|----------|----------|------|--------|
|                             | 0.050m    | 0       | 0   | 0      | 0        | 0        | 0        | Hard | OK     |

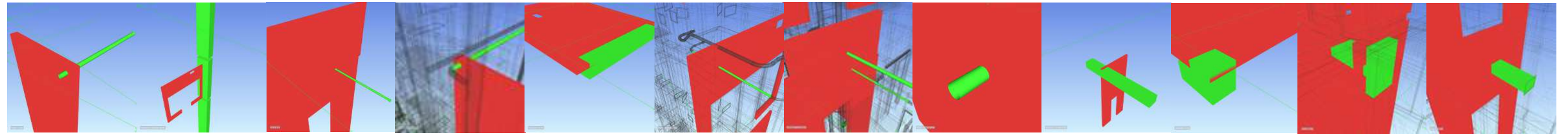
| T04_[MEP]-Set01-[MEP]-Set01 | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type                | Status |
|-----------------------------|-----------|---------|-----|--------|----------|----------|----------|---------------------|--------|
|                             | 0.020m    | 374     | 0   | 374    | 0        | 0        | 0        | Hard (Conservative) | OK     |

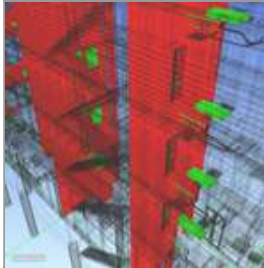
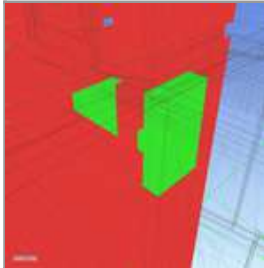
| T02_[ARC]-Set01-[MEP]-Set01 | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type | Status |
|-----------------------------|-----------|---------|-----|--------|----------|----------|----------|------|--------|
|                             | 0.050m    | 129     | 0   | 129    | 0        | 0        | 0        | Hard | OK     |



# FINAL CLASH DETECTION

|                             |           |         |     |        |          |          |          |                     |        |
|-----------------------------|-----------|---------|-----|--------|----------|----------|----------|---------------------|--------|
| T01_[ARC]-Set01-[STR]-Set01 | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type                | Status |
|                             | 0.050m    | 24      | 0   | 0      | 24       | 0        | 0        | Hard                | OK     |
| T02_[ARC]-Set01-[MEP]-Set01 | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type                | Status |
|                             | 0.050m    | 62      | 0   | 0      | 62       | 0        | 0        | Hard                | OK     |
| T03_[STR]-Set01-[MEP]-Set01 | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type                | Status |
|                             | 0.050m    | 0       | 0   | 0      | 0        | 0        | 0        | Hard                | OK     |
| T04_[MEP]-Set01-[MEP]-Set01 | Tolerance | Clashes | New | Active | Reviewed | Approved | Resolved | Type                | Status |
|                             | 0.020m    | 374     | 0   | 374    | 0        | 0        | 0        | Hard (Conservative) | OK     |

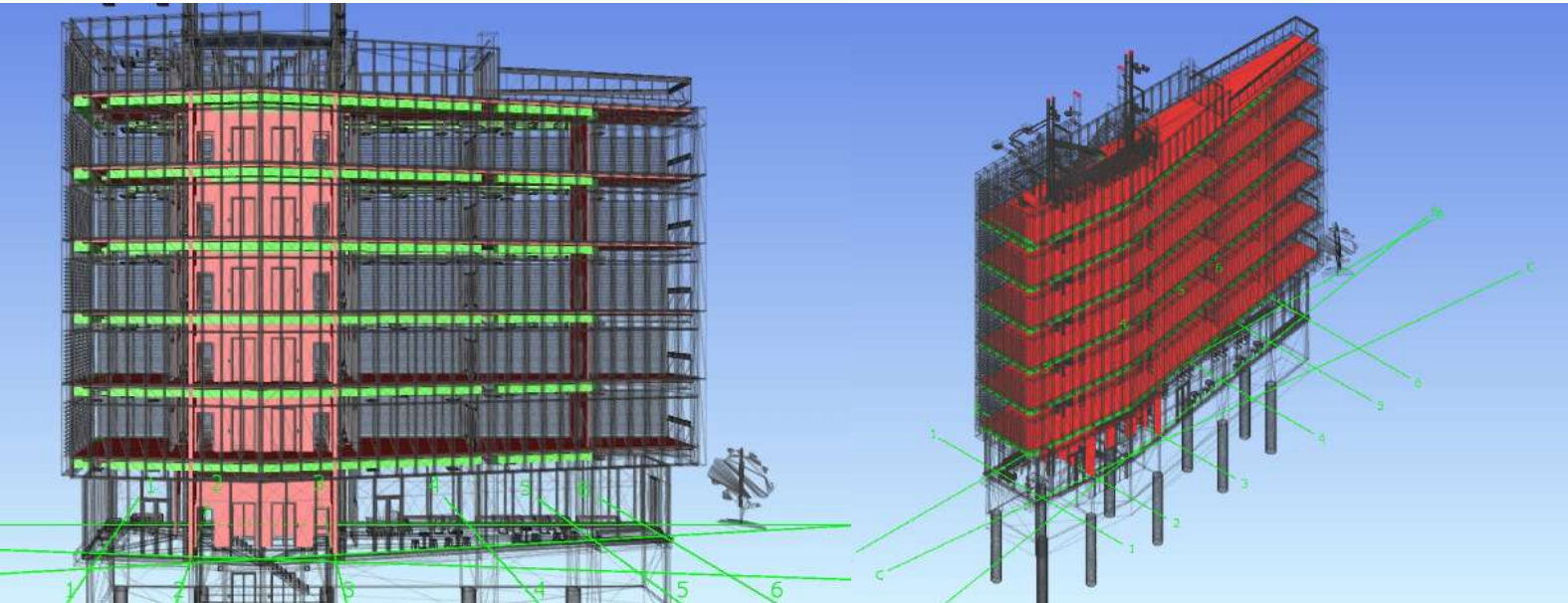


|                                                                                    |          |        |           |      |                 |       |                               |                    |              |       |                    |          |       |
|------------------------------------------------------------------------------------|----------|--------|-----------|------|-----------------|-------|-------------------------------|--------------------|--------------|-------|--------------------|----------|-------|
|  | Clash2   | Active | B-3 : L 0 | Hard | 2024/2/18 23:48 | Vivek | x:-17.025, y:-5.125, z:8.766  | Element ID: 747385 | Default Wall | Solid | Element ID: 312877 | Paint-09 | Solid |
|  | Clash20  | Active | B-3 : L 5 | Hard | 2024/2/18 23:48 | Vivek | x:-15.625, y:-5.258, z:27.140 | Element ID: 876031 | Default Wall | Solid | Element ID: 312877 | Paint-09 | Solid |
|  | Clash103 | Active | A-3 : L 0 | Hard | 2024/2/18 23:48 | Vivek | x:-14.645, y:-5.727, z:8.778  | Element ID: 748464 | Default Wall | Solid | Element ID: 312883 | Paint-09 | Solid |

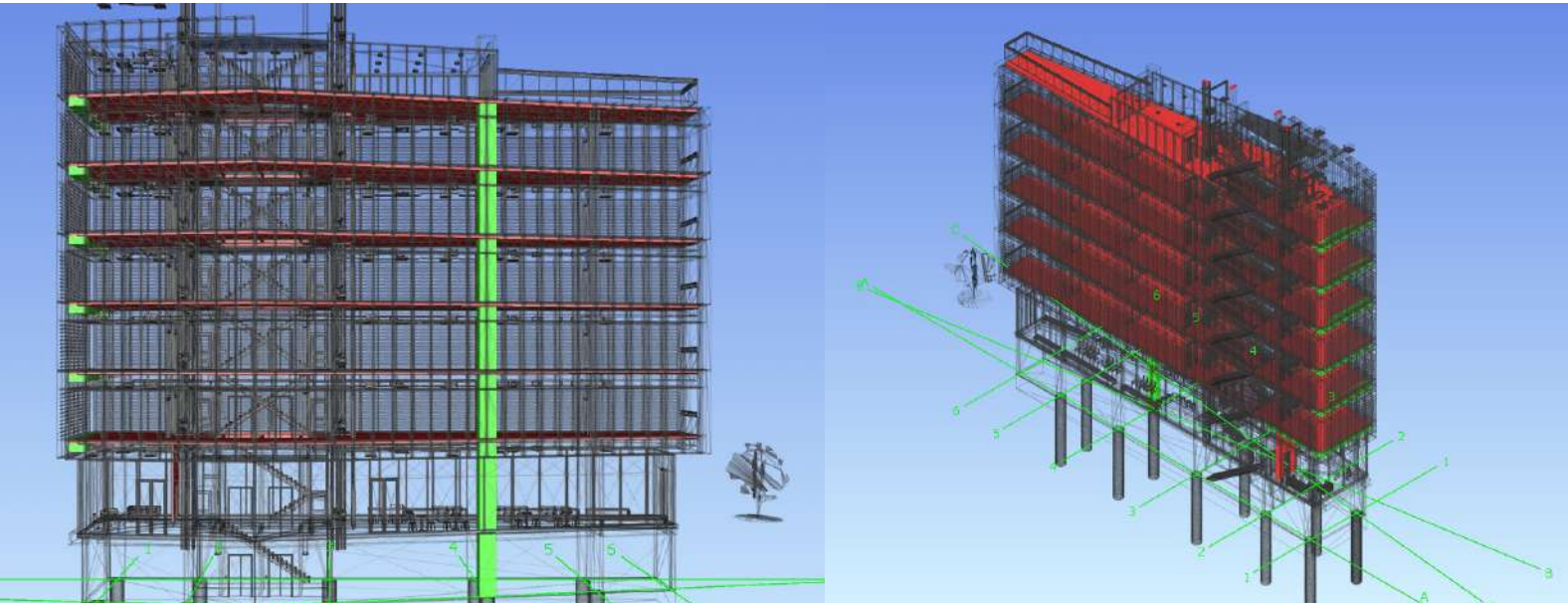


# CLASH RESOLUTION COMPARISON

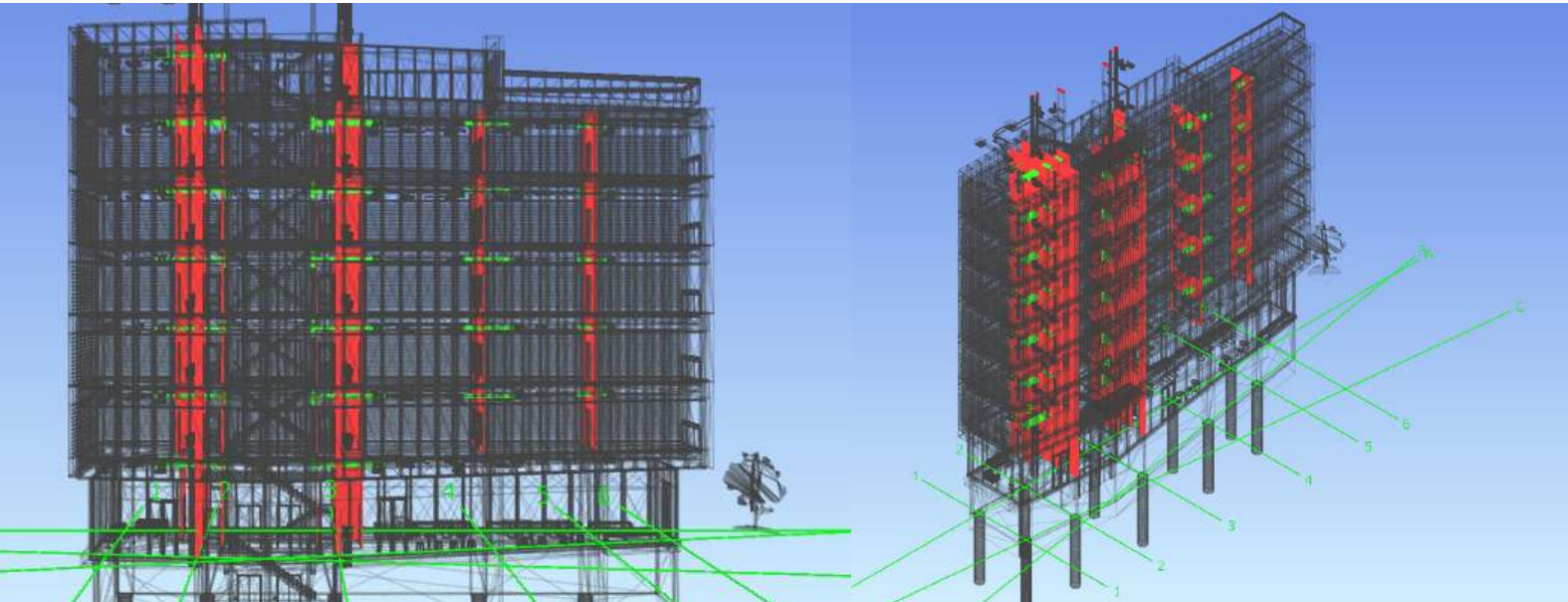
| T01_[ARC]-Set01-[STR]-Set01 | Set     | Clashes | Type | Status |
|-----------------------------|---------|---------|------|--------|
|                             | Initial | 76      | Hard | OK     |



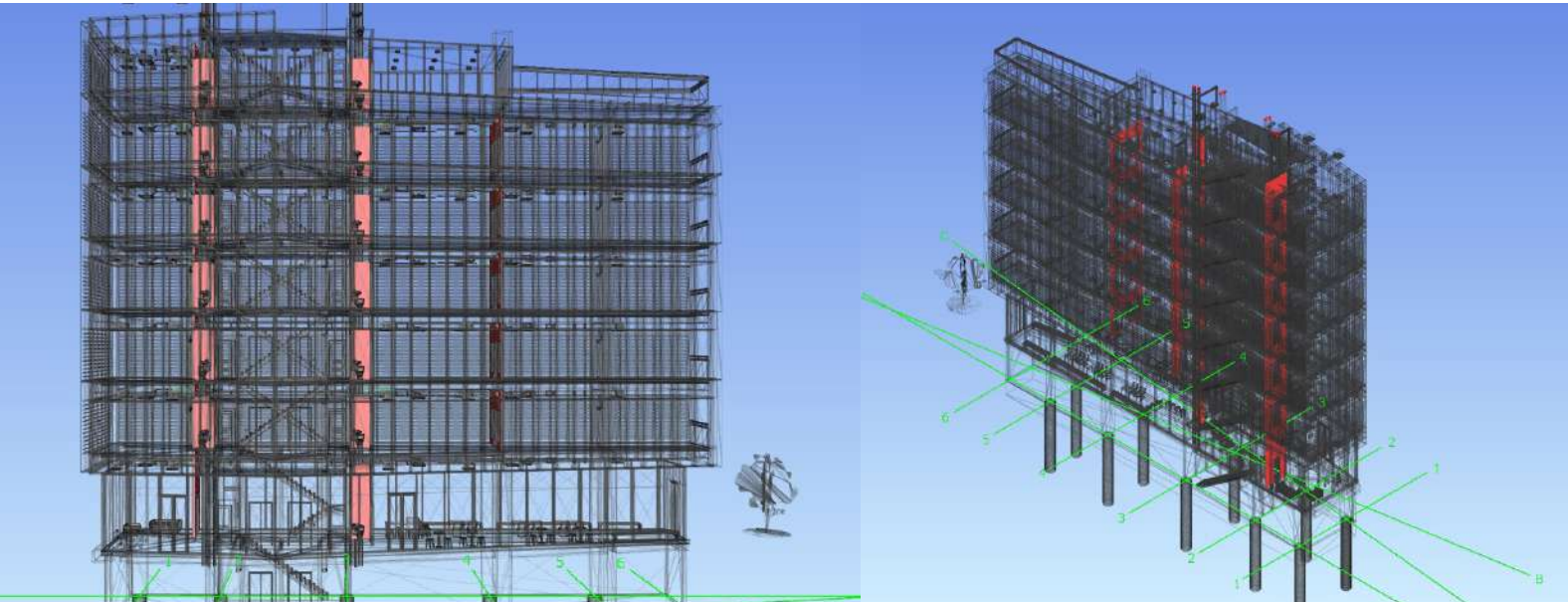
| T01_[ARC]-Set01-[STR]-Set01 | Set     | Clashes | Type | Status |
|-----------------------------|---------|---------|------|--------|
|                             | Initial | 76      | Hard | OK     |



| T02_[ARC]-Set01-[MEP]-Set01 | Set     | Clashes | Type | Status |
|-----------------------------|---------|---------|------|--------|
|                             | Initial | 129     | Hard | OK     |



| T02_[ARC]-Set01-[MEP]-Set01 | Set   | Clashes | Type | Status |
|-----------------------------|-------|---------|------|--------|
|                             | Final | 62      | Hard | OK     |





# 4D AND 5D SIMULATIONS

New GroupNew ItemUse ResourceDeleteProperty Mapping

ItemsWBS

|                        |     |
|------------------------|-----|
| Concrete structure     | 1   |
| Column                 | 1.1 |
| Beam                   | 1.2 |
| Structural Wall        | 1.3 |
| Structural slab        | 1.4 |
| Staircase structure    | 1.5 |
| Architectural Elements | 2   |
| Facade                 | 2.1 |
| Curtain wall           | 2.2 |
| Doors                  | 2.3 |
| Floor Finish           | 2.4 |
| Partition walls        | 2.5 |
| Railings               | 2.6 |
| Staircase Finish       | 2.7 |

Item NameWork Breakdown Structure

Doors3

Description

Object Appearance

ColorOpaqueTransparentLine Thickness0.3281 ft

Item CalculationsItem Map Rules

| Variable  | Formula         | Units  |
|-----------|-----------------|--------|
| Length    | =ModelLength    | Meters |
| Width     | =ModelWidth     | Meters |
| Thickness | =ModelThickness | Meters |
| Height    | =ModelHeight    | Meters |

Double-click a formula to edit it inline.

Quantification Workbook

Model TakeoffVirtual TakeoffSelectHide TakeoffShow TakeofffxDoorsChange AnalysisUpdate

Select MarkupFilter Markup

ItemsWBS

|                        |     |
|------------------------|-----|
| Concrete structure     | 1   |
| Column(6)              | 1.1 |
| Beam(131)              | 1.2 |
| Structural Wall(81)    | 1.3 |
| Structural slab(8)     | 1.4 |
| Staircase structure(4) | 1.5 |
| Architectural Elements | 2   |
| Facade(0)              | 2.1 |
| Curtain wall(3406)     | 2.2 |
| Doors(75)              | 2.3 |
| Floor Finish(9)        | 2.4 |
| Partition walls(92)    | 2.5 |
| Railings(6)            | 2.6 |
| Staircase Finish(4)    | 2.7 |

| Status | WBS/RBS | Name  | Length  | Width    | Thickness | Height    |
|--------|---------|-------|---------|----------|-----------|-----------|
|        | 2.3     | Doors | 0.000 m | 60.538 m | 3.399 m   | 161.771 m |

| Status | WBS   | Object                 | ModelLength | ModelWidth | ModelThickness | ModelHeight |
|--------|-------|------------------------|-------------|------------|----------------|-------------|
|        | 2.3.1 | Vivek Khanna Fire Door |             | 1.000 m    | 0.050 m        | 2.000 m     |
|        | 2.3.2 | DOOR_D1, D2, D3        |             | 0.900 m    | 0.051 m        | 2.100 m     |
|        | 2.3.3 | DOOR_D1, D2, D3        |             | 0.900 m    | 0.051 m        | 2.100 m     |
|        | 2.3.4 | DOOR_D1, D2, D3        |             | 0.900 m    | 0.051 m        | 2.100 m     |
|        | 2.3.5 | DOOR_D1, D2, D3        |             | 0.900 m    | 0.051 m        | 2.100 m     |
|        | 2.3.6 | DOOR_D1, D2, D3        |             | 0.900 m    | 0.051 m        | 2.100 m     |
|        | 2.3.7 | DOOR_D1, D2, D3        |             | 0.900 m    | 0.051 m        | 2.100 m     |
|        | 2.3.8 | DOOR_D1, D2, D3        |             | 0.900 m    | 0.051 m        | 2.100 m     |

Quantification WorkbookResource Catalog

